Internet Appendix

Appendix A The Ecuadorian Banking Sector

Over our sample, from 2010 to 2017, the Ecuadorian financial system included 33 lenders. The Superintendencia de Bancos y Seguros (SB; Superintendent of Banks and Insurance Companies) is the regulator for the sector.[1]

Public financial institutions include Banco del Estado, Banco Ecuatoriano de la Vivienda, Banco Nacional de Fomento (later BanEcuador), and Corporación Financiera Nacional. There are also 29 private lenders operating over our period. Of these, Banco del Pacífico has been government-owned since a 1999 banking crisis, but is regulated as a private bank by SB. It is a large, national lender. Citibank is a foreign-owned subsidiary with a private bank charter, unlike, for example, Lloyds TSB Bank and Barclays, whose loans do not enter our sample as they have foreign bank charters. Four private banks (Finca, D-Miro, VisionFund, and CoopNacional) are primarily microlenders, with regular commercial loans making up less than 1% of their loan portfolio.

The remaining 23 private banks include three other large national lenders, based on total assets (Pichincha, Guayaquil, and Produbanco), eight medium-sized banks (Bolivariano, Internacional, Austro, General Rumiñahui, Machala, Loja, Solidario and Procredit), and twelve smaller banks (Amazonas, Cofiec, Comercial de Manabí, Del Litoral, Sudamericano, Territorial, Unibanco, Capital, BP Diners, Proamerica, Delbank, and BancoDesarrollo).

Interest rates on new credits are regulated by a body under the control of the legislature, the Junta de Política y Regulación Monetaria y Financiera. It defines maximum interest rates for credit segments. For commercial credit, maximum interest rates vary according to the size of the loan and the size of the company.[2] Finally, depositors are protected by deposit insurance from the Corporación del Seguro de Depósitos (Deposit Insurance Corporation (COSEDE)).

Appendix A.1 Cross-country benchmarking

Overall, Ecuador is typical of many middle-income and bank-dependent economies studied in the literature. We document this by benchmarking Ecuador against other Latin American and Caribbean (LAC) countries and against the emerging market economy (EME) average over 2010-2017. We use the World Bank's Global Financial Development Database (GFDD), which compiles comparable banking-sector indicators for over 200 economies from national regulatory sources.

[1] This does not include savings and credit cooperatives (cooperativas de ahorro y crédito), which are regulated by the Superintendencia de Economía Popular y Solidaria (Superintendent of the Popular and Solidarity Economy). In our study we focus on the regular commercial lending sector.

[2] Interest rate caps are common around the world. As of 2018, approximately 76 countries (representing 80% of world GDP) impose some restrictions on interest rates, according to the World Bank. They are particularly prevalent in Latin America and the Caribbean but are also observed on some financial products offered in Australia, Canada and the United States (see Ferrari et al. (2018)). Interest rate caps place constraints on bank market power and affect the distribution of credit. This is reflected in our model.

First, Figure A1 compares Ecuador across a set of bank indicators capturing cost efficiency, profitability, and asset quality. The cost-to-income ratio (operating costs over operating income) measures how efficiently banks convert revenue into profit, with higher values indicating lower efficiency. Overhead costs as a share of assets captures the weight of non-interest expenses (salaries, infrastructure) relative to bank size. Return on equity (net income over shareholder equity) measures bank profitability. The non-performing loan (NPL) ratio (loans in default over total gross loans) captures credit risk and loan-portfolio quality. The red lines mark the average for other (non-LAC) emerging-market economies. Ecuador's position varies across these dimensions but remains within the range of its LAC peers and the EME ex LAC average over our sample period.

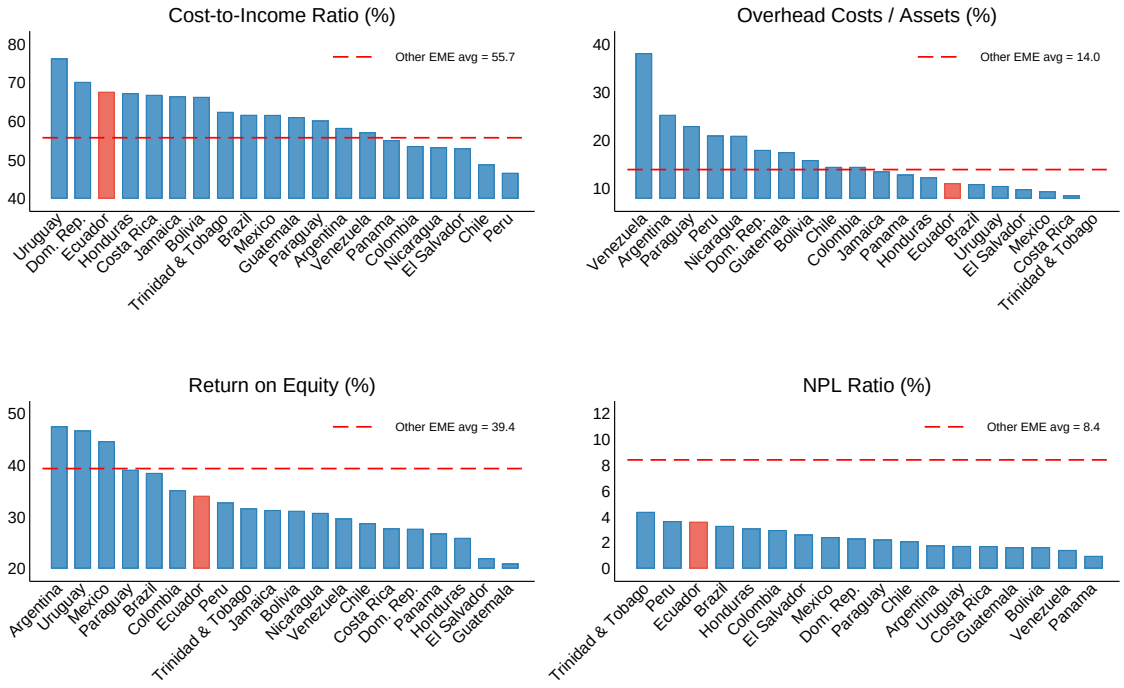


FIGURE A1: BANKING SECTOR INDICATORS ACROSS LAC COUNTRIES (2010-2017 AVERAGE)

Each panel ranks LAC countries by a banking-sector indicator averaged over 2010-2017. Ecuador is highlighted in red. The dashed red line marks the average for other (non-LAC) emerging-market economies. Panels show (a) cost-to-income ratio (operating costs/operating income), (b) overhead costs as a share of total assets, (c) return on equity (net income/shareholder equity), and (d) non-performing loan ratio (loans in default/total gross loans). Source: World Bank GFDD.

Second, Figure A2 compares Ecuador to its regional peers and the EME ex LAC average on two measures related to the market structure of the banking system. In the left panel, the net interest margin (NIM) is the gap between what banks earn on loans and pay on de-

posits, expressed as a share of interest-bearing assets. It reflects the price of intermediation that borrowers face and is shaped by competitive conditions, risk, and operating costs. Ecuador's average NIM of 6.4% places it near the regional median.

In the right panel we plot Ecuador's share of total banking assets held by the three largest banks alongside the other LAC countries. This is a common proxy for how market power is distributed across lenders. In Ecuador it is 73.9%, near the middle of the LAC distribution. This reflects the common market structure of a core group of large banks and a wider fringe of smaller banks.

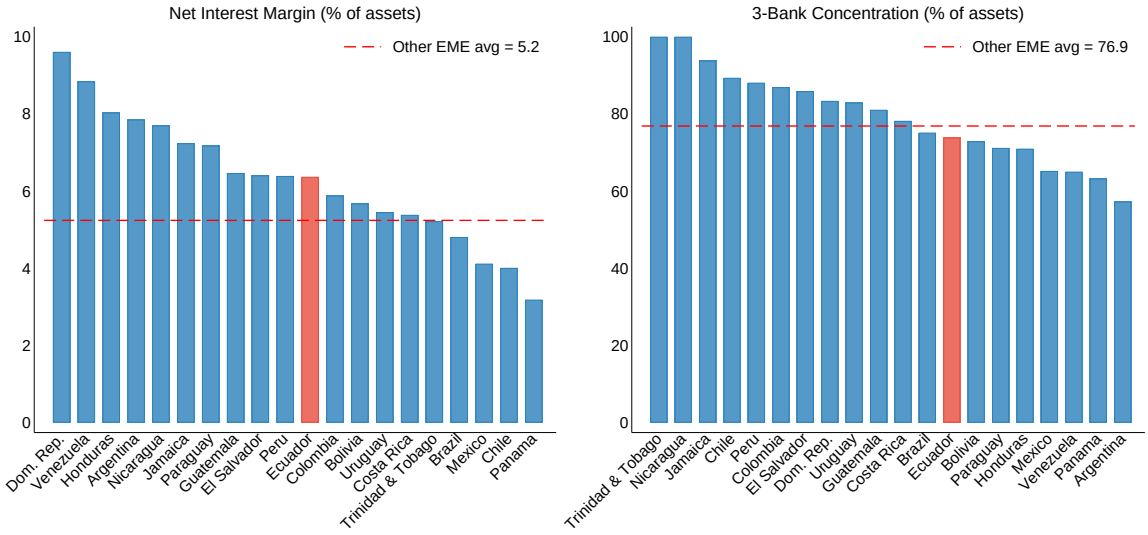


FIGURE A2: NET INTEREST MARGINS AND THREE-BANK CONCENTRATION ACROSS LAC COUNTRIES (2010-2017 AVERAGE)

Each bar shows the 2010-2017 country-level average. Ecuador is highlighted in red. The solid red line marks the average for other (non-LAC) emerging-market economies. Source: World Bank GFDD.

Finally, Figure A3 shows that high three-bank concentration is not specific to LAC. Every major world region has an average three-bank concentration ratio above 60% except in South Asia, and Ecuador's ratio falls in the middle of this global distribution.

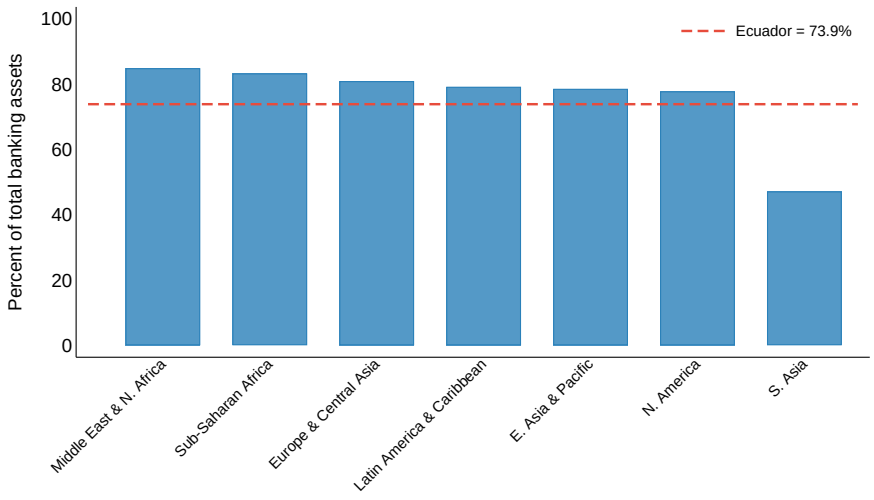


FIGURE A3: THREE-BANK CONCENTRATION BY WORLD REGION (2010-2017 AVERAGE)

Each bar shows the average three-bank asset concentration ratio for a world region over 2010-2017, computed as the equal-weighted mean of country-level averages. The dashed red line marks Ecuador. Source: World Bank GFDD.

Overall, Ecuador is broadly representative of its regional peers and other EME economies.

Appendix A.2 Bank liquidity and reserve adequacy

Banks in our sample maintained ample liquid asset holdings and were not funding constrained around the introduction of the SOLCA tax. Figure A4 plots the cross-bank distribution of two key liquidity ratios at the monthly level: the ratio of liquid assets to short-term deposits (Panel a) and to total assets (Panel b). Both ratios are stable, with no visible deterioration around the introduction of the SOLCA tax in October 2014 (dashed vertical line). The median bank held liquid assets equal to approximately 25-30% of short-term deposits and 15-20% of total assets, and these levels were maintained throughout the post-tax period.

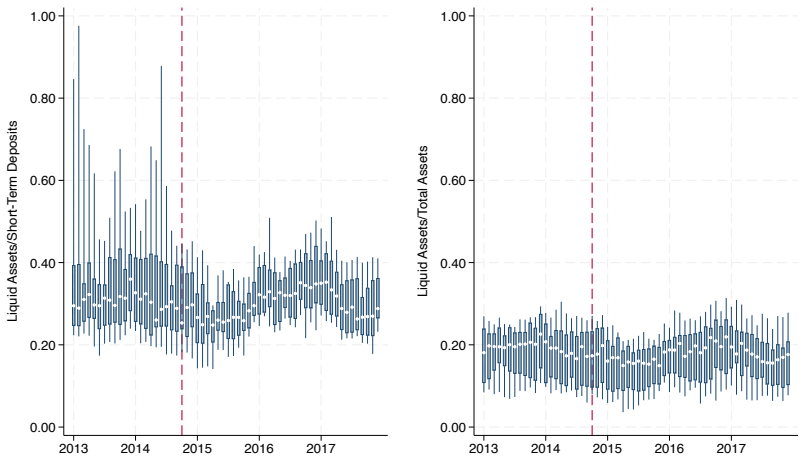


FIGURE A4: BANK LIQUIDITY RATIOS AROUND THE SOLCA TAX

The figure reports the cross-bank distribution of two liquidity ratios at the monthly level. Panel (a) plots the ratio of liquid assets to total short-term deposits; Panel (b) plots the ratio of liquid assets to total assets. Each monthly box plot shows the median (white dot), interquartile range (box), and 10th-90th percentile range (whiskers) across banks. The dashed vertical line marks the introduction of the SOLCA tax in October 2014. Source: Superintendencia de Bancos, monthly balance sheet data.

Appendix A.3 Market characteristics' relationship to interest rates

We test the representativeness of Ecuadorian commercial lending by checking the correlations between average equilibrium interest rates and market characteristics at the bank-province-year level. Table A1 reports the results. Model 1 employs year fixed effects (FE), Model 2 utilizes province and year FE, and Model 3 runs estimates with both year and bank FE.

TABLE A1: INTEREST RATE AND MARKET CHARACTERISTICS

The table reports correlations between average nominal interest rates on new commercial credit and market characteristics. Data are at the bank-province-year level for years in which the bank offered any loan in a given province. *Av. Interest Rate* is the nominal, annualized interest rate at issuance, in percent. The variables include the natural log transformation of: *Av. Loan* is the average loan size at issuance; *Av. Maturity* is average annualized term-to-maturity at issuance; # *Branches* is the number of open loan-granting branches in the province; # *Other Branches* is the total number of competing loan-granting branches active in the province; *HHI* is the Herfindahl-Hirschman Index at the province-year level based on commercial lending share; # *Loans per Client* is the average number of loans extended per firm from a given bank; *# Clients* is the sum of unique clients. Data from state-owned banks are excluded. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Variable	**(1)** **Av. IR**	**(2)** **Av. IR**	**(3)** **Av. IR**
Ln(Av. Loan)	-0.568***	-0.606***	-0.558***
	(0.045)	(0.047)	(0.053)
Ln(Av. Maturity)	-0.615***	-0.582***	-0.543**
	(0.186)	(0.195)	(0.226)
Ln(# Branches)	-0.419***	-0.409***	-0.344**
	(0.141)	(0.142)	(0.156)
Ln(# Other Branches)	-0.062	0.052	-0.000
	(0.059)	(0.082)	(0.086)
Ln(HHI Value)	0.682***	0.539	0.333
	(0.213)	(0.366)	(0.213)
Ln(# Loans per Client)	-0.606***	-0.606***	-0.475***
	(0.048)	(0.048)	(0.053)
Ln(# Clients)	0.508***	0.581***	0.271***
	(0.052)	(0.063)	(0.052)
Constant	12.219***	13.116***	14.852***
	(1.895)	(2.932)	(1.907)
Year FE	Yes	Yes	Yes
Province FE	No	Yes	No
Bank FE	No	No	Yes
Observations	1,734	1,734	1,734
R-squared	0.297	0.344	0.415

The general patterns we observe between market access and loan pricing align with those documented in existing literature in Latin America and elsewhere. Across all our models, we find that average interest rates tend to decline with increasing loan size and maturity. Banks that have a higher number of loan-granting branches in a given market on average offer lower rates, potentially indicating that banks expand in markets in which they have an efficiency advantage. Conversely, we find a weak and statistically insignificant link between loan pricing and the number of competing branches within a province or across different markets served by the same bank. This suggests that mere access to competing banks through larger branches does not significantly influence a bank's average pricing strategy.

Moreover, we uncover a positive correlation between market concentration, as proxied by the Herfindahl-Hirschman Index (HHI) based on commercial lending share, and average

interest rates. This association between less competitive empirical competition proxies and higher loan rates or lower deposit rates is consistent with findings for U.S. commercial lending (Berger and Hannan (1989); Hannan (1991)), European banking (Degryse and Ongena (2005); Van Leuvensteijn et al. (2013)), and cross-country samples (Beck et al. (2013)). With bank fixed effects, the concentration coefficient remains positive but is statistically insignificant.

Interest rates tend to be lower when the bank and borrower interact frequently, as measured by the number of loans per borrower. An extensive relationship lending literature documents a negative correlation between relationship strength and loan rates (Petersen and Rajan (1995); Degryse and Ongena (2005)). However, larger banks (as indicated by the number of borrowers) generally charge higher interest rates. This could be due to the diverse needs (borrower preference heterogeneity) that lead firms to borrow from specific banks, despite steeper prices.

Appendix A.4 Loan default in our data relative to the literature

In our dataset of commercial loans to non-micro, formal firms, we observe low realized default of less than one percent, on average. Consistent with this, the banks in our sample only write down the value of about two percent of the loans that they issue. In the total sample, which includes sole proprietorships and micro-loans, the default rate is three percent. Here, we benchmark against default in related papers:

- Crawford et al. (2018) report a default rate of 6% in a sample of Italian small business lines of credit (with maturity 6 months to a year) between 1988 and 1998, which included a financial crisis in 1992.
- Ioannidou et al. (2022) report between 4 to 12% of loans flagged as risky by banks and effective default between 12 to 26% in Bolivia.

Low default in our sample is also consistent with broader regional patterns. World Bank GFDD data show that aggregate non-performing loan ratios in LAC countries averaged 2–4% over 2010–2017, well below the 8.4% average for other emerging-market economies (Figure A1, panel d). And these aggregate ratios include riskier consumer and micro-enterprise lending; whereas commercial loans to established formal firms tend to be much lower.

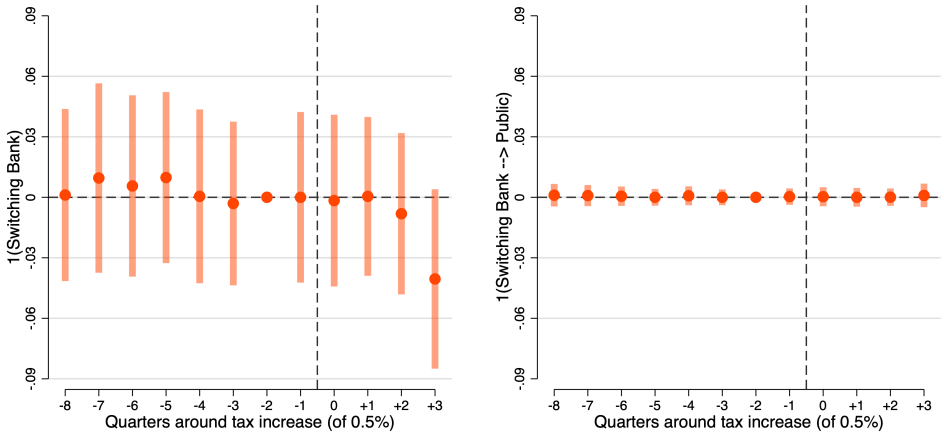
Appendix A.5 Commercial lending of private and public banks

The government banks specialize in the commercial loan market in lending to small firms in small markets, and to large exporters. On average (at the median) they lend 20.2% (10.5%) of the outstanding commercial debt in a given province-year, or 8.8% when the average is weighted by market size. At the borrower-year-level they lend 2.3% (0%) on average (at the median). Thus, there is some degree of competition between the public and private banking sector in commercial lending and there are possible indirect effects of the SOLCA tax on public commercial lending. For this reason, we include the public banks in the model estimation.

While theoretically salient, the existence of the public commercial loan market does not appear to be first order in practice in this setting. This is suggested by Figure 2, where we see no

reaction in interest rates to the introduction of the SOLCA tax, and confirmed in Figure B2 and Table B1, which treat public loans as the control group in a difference-in-differences analysis. In unreported results, there was no significant effect on loan maturity or amount borrowed in loans lent by public banks.

We also see no evidence that there was significant switching of firms around the introduction of the tax, either between any banks or from borrowing taxed private-sector loans to borrowing untaxed public-sector loans. To test this, we first define the variable *Switch*, which takes the value one if the loan borrowed in period t is from a different bank, public or private, than the last loan borrowed. The left-hand panel of Figure A5 reports the evolution in the probability of switching lenders around the introduction of the SOLCA tax relative to the probability two quarters before the tax was introduced. We see no significant difference either leading up to the tax or immediately after its introduction. If anything, there is a decrease in the probability of switching lenders three quarters after the introduction of the tax, though it is not significant at conventional confidence levels. This may reflect the macroeconomic shock from falling oil prices that sent Ecuador into recession in the first quarter of 2015. In the right-hand panel of Figure A5 we similarly see no evidence of a change in the probability of switching to borrowing from a public bank.

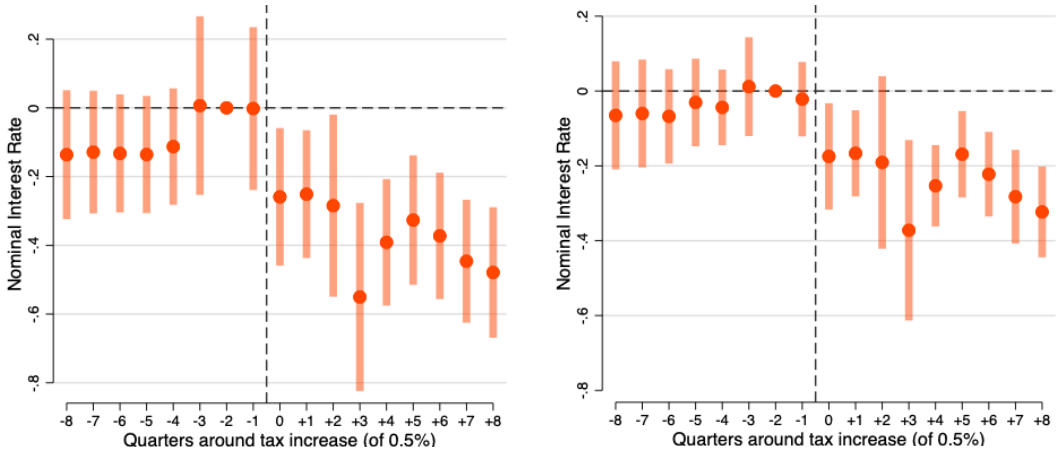


(a) Prob. switch bank (b) Prob. switch to public bank

FIGURE A5: DYNAMIC ANALYSIS OF THE INTRODUCTION OF THE SOLCA TAX ON THE PROBABILITY OF SWITCHING LENDERS

The figure reports the period-by-period difference in outcomes around the introduction of the SOLCA tax relative to event-time period $t = -2$ (normalized to zero). The outcome in Figure (a) is the probability that a new commercial loan is borrowed from a different bank than the last loan in the period relative to at $t = -2$. For Figure (b) the outcome is the probability that a new loan is borrowed from a different bank than the last loan and that new lender is a public, state-owned bank. Data are loan-level and include bank fixed effects. Standard error bars are shown at the 95% confidence level and are clustered at the bank-quarter level.

Appendix B Robustness of pass-through estimates



(a) Regular loans; bank & firm FE (b) Regular loans, bank-firm pair FE

FIGURE B1: DYNAMIC ANALYSIS OF THE SOLCA TAX ON PRE-TAX NOMINAL INTEREST RATES OF NEW COMMERCIAL CREDIT BY PRIVATE BANKS: EIGHT-QUARTER WINDOW

This figure replicates Figure 1 using an event-study window spanning eight quarters before and after the SOLCA tax implementation. It reports quarter-specific differences in average pre-tax nominal interest rates on new regular commercial loans from private banks, relative to event time $t = -2$ (the omitted period). Panel (a) includes bank and firm fixed effects; Panel (b) includes bank–firm pair fixed effects. Both specifications include loan predicted default at origination and flexible controls for loan maturity and amount. The data are at the loan level. Bars show 95% confidence intervals based on standard errors clustered at the bank-quarter level.

TABLE B1: PRIVATE VERSUS PUBLIC PASS-THROUGH ESTIMATES

This table reports difference-in-differences estimates comparing pass-through of the SOLCA loan tax to interest rates on commercial loans by private banks (treated) versus government-owned public banks exempt from the tax (control). Data are at the loan level, from eight quarters before through three quarters after the October 2014 introduction, excluding October 2014. The dependent variable is the nominal annualized interest rate in percentage points. *Post* equals one for quarters after the tax; *Treated* equals one for banks whose loans became taxable. Regressions control for twenty buckets each of term to maturity and loan amount. Standard errors clustered at the bank-quarter level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)
Post × Treated	−0.214**	−0.208***
	(0.084)	(0.070)
Pair FE	Yes	No
Firm FE	No	Yes
Bank FE	No	Yes
Quarter FE	Yes	Yes
Maturity & amount controls	Yes	Yes
Adjusted R^2	0.769	0.710
Observations	400,227	406,943

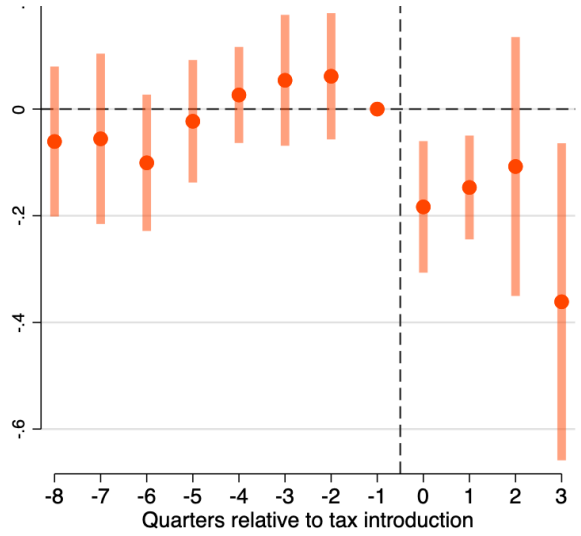
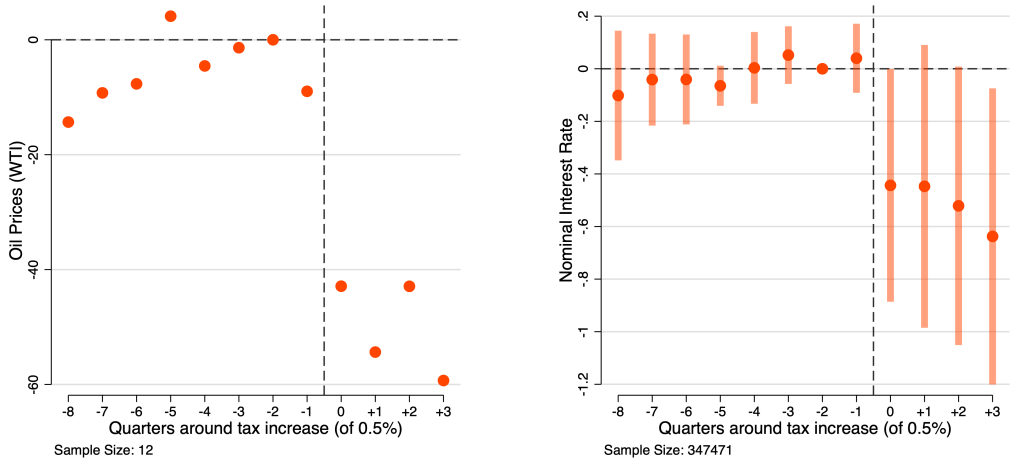


FIGURE B2: DYNAMIC ANALYSIS OF THE SOLCA TAX ON THE INTEREST RATE ON NEW COMMERCIAL DEBT LENT BY PRIVATE VS. PUBLIC BANKS

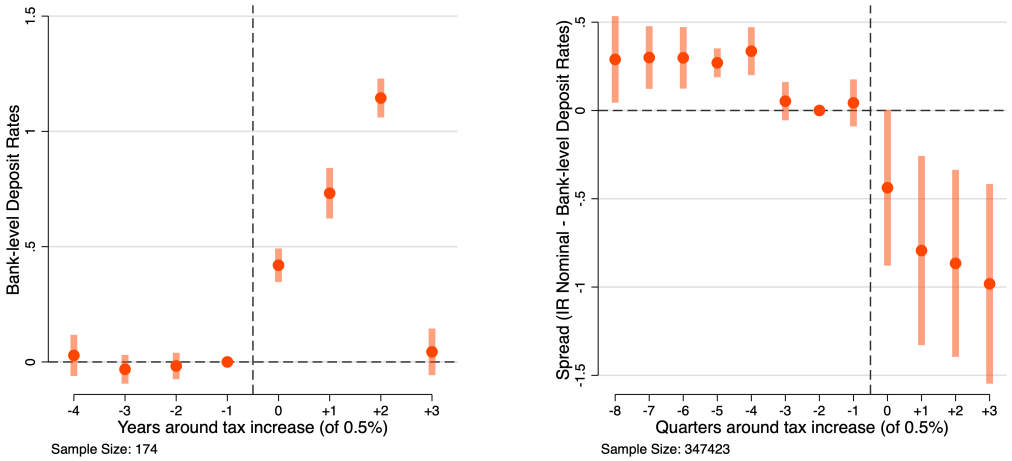
The figure reports period-by-period difference-in-differences coefficients from eight quarters before through three quarters after the introduction of the SOLCA tax. Loans granted in October 2014, represented by the dashed vertical line, are removed. Each coefficient is from the interaction of a quarter indicator with *Treated*, which equals one for loans issued by private banks subject to the tax and zero for loans issued by government-owned public banks exempt from the tax, with the quarter immediately preceding the reform ($t = -1$) as the omitted reference period. Data are at the loan level for new commercial loans. Coefficients are estimated with bank-firm pair fixed effects. Standard error bars are shown at the 95% confidence level, based on standard errors clustered at the bank-quarter level.



(a) Oil Prices (b) Interest Rates cont. Oil

FIGURE B3: ROBUSTNESS CONTROLLING FOR OIL PRICES

Panel (a) reports the evolution of quarterly prices of WTI Crude. Panel (b) reports the period-by-period difference in average pre-tax nominal interest rate on new commercial loans from private banks around treatment assignment, controlling for buckets of amount and maturity, default probability, a third-degree polynomial of quarterly WTI Crude prices, and bank-firm pair FE. Standard error bars are shown at the 95% confidence level and are clustered at the bank-quarter level. The relative event-time period $t = -2$ is normalized to zero.



(a) Deposit Rates (b) Spread

FIGURE B4: ROBUSTNESS ON SPREADS

Panel (a) reports the evolution of yearly deposit rates at the bank level around 2014 using bank FE. Yearly deposit rates are calculated by obtaining national average deposit rates by product and calculating bank-level yearly averages across products using product portfolio bank shares. Panel (b) reports the period-by-period difference in average spreads (pre-tax nominal transaction-level interest rate minus yearly bank-level deposit rate average) on new commercial loans from private banks around treatment assignment controlling for buckets of amount and maturity, default probability, a third-degree polynomial of quarterly WTI Crude prices, and bank-firm pair FE. Standard error bars are shown at the 95% confidence level and are clustered at the bank-level in Panel (a) and at the bank-quarter level in Panel (b).

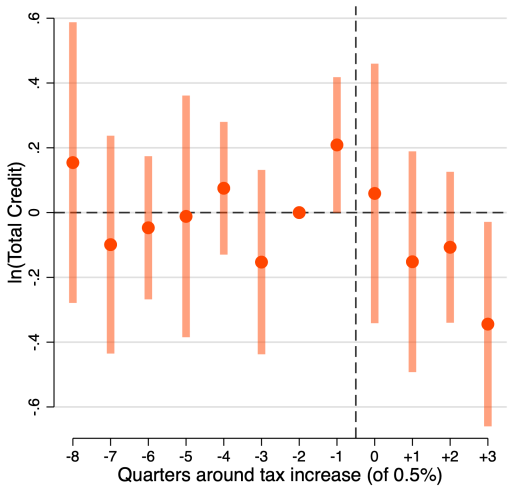
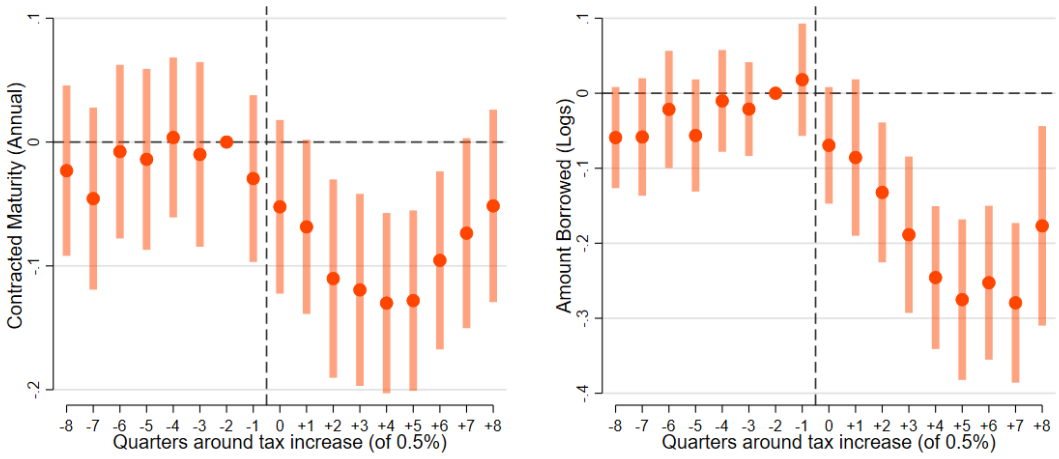


FIGURE B5: DYNAMIC ANALYSIS OF THE INTRODUCTION OF THE SOLCA TAX ON TOTAL COMMERCIAL CREDIT LENT BY PRIVATE BANKS

The figure reports the period-by-period difference in total commercial lending from private banks around treatment assignment relative to event-time period $t = -2$ (normalized to zero), using bank fixed effects. Data are bank-quarter level on commercial loans granted by private banks to Ecuadorian corporations. Standard errors bars are shown at the 95% confidence level and are clustered at the bank level.



(a) Regular loans, maturity; bank & firm FE (b) Regular loans, amount, bank & firm FE

FIGURE B6: DYNAMIC ANALYSIS OF THE SOLCA TAX ON MATURITY AND AMOUNT OF NEW COMMERCIAL DEBT LENT BY PRIVATE BANKS

Replicating Figure 3 in an eight quarter estimation around the introduction of the SOLCA tax. It reports the period-by-period difference in average term-to-maturity (Panel (a)) or the natural logarithm of the amount borrowed (Panel (b)) on new commercial loans from private banks around treatment assignment relative to event-time period $t = -2$ (normalized to zero), using firm $\times$ bank fixed effects. Data are loan-level. Standard error bars are shown at the 95% confidence level and are clustered at the bank-quarter level.

TABLE B2: ROBUSTNESS OF PASS-THROUGH ESTIMATES

The table reports aggregate pass-through estimates to the tax-inclusive interest rates of commercial loans around the introduction of the 2014 SOLCA tax in Ecuador. Data are at the loan-level for 2010–2017, excluding October 2014. The main independent variable is the tax rate, measured as 0.5 adjusted proportionally by term-to-maturities if maturity is less than 1 year. The dependent variable is the tax-inclusive interest rate, which is the sum of the nominal, annualized interest rate plus the tax rate, in percentage points. Regressions control for twenty buckets of term-to-maturity, twenty buckets of loan amount, and the loan's predicted default probability. Column (1) additionally controls for a third-degree polynomial of quarterly prices of WTI crude oil. Column (2) controls for a third-degree polynomial of province quarterly GDP growth. Column (3) includes quarter fixed effects. Robust standard errors clustered at the bank-quarter level are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. The table bottom of the table reports testing against the full pass-through null hypothesis ($PT_o = 1$).

	Outcome: Tax-inclusive interest rate		
	(1)	(2)	(3)
Pass-through ($\widehat{PT}$)	0.613	0.403*	0.702*
	(0.388)	(0.214)	(0.371)
WTI Oil Price	0.129***		
	(0.043)		
(WTI Oil Price)^2	-0.002***		
	(0.001)		
(WTI Oil Price)^3	0.000***		
	(0.000)		
Province GDP Growth		-0.014	
		(0.026)	
(Province GDP Growth)^2		-0.003	
		(0.006)	
(Province GDP Growth)^3		-0.000	
		(0.004)	
Pr(Default) Control	Yes	Yes	Yes
Maturity & Amount Controls	Yes	Yes	Yes
Pair Fixed Effect	Yes	Yes	Yes
Year-quarter Fixed Effect	No	No	Yes
Observations	347,471	489,251	489,251
R-squared	0.777	0.749	0.749

Appendix C Price and Default Prediction

Appendix C.1 Price Variation

A central empirical challenge in estimating our model is that we observe loan terms only for granted loans, whereas the demand estimation requires the counterfactual prices that all available banks would offer to every potential borrower. To address this, we construct discrete pricing bins conditioned on observable firm characteristics. We classify borrowers into categories defined by: (1) Credit Purpose = {General Loan, Other}, (2) Segment = {Corporate, Enterprise, SMEs, Other Commercial}, (3) Risk = {Risky (Risk weight < B), Non-Risky}, and (4) Maturity (divided into four buckets). The interaction of these categories defines a borrower group g. For each region and group g we calculate the average price offered by banks active in those segments.

This discretization captures price variation effectively, as shown in Table C1. The table reports adjusted R^2 for different specifications regressing loan-level, pre-tax nominal interest rates on various controls. The first row regresses loan-level pre-tax nominal interest rates on individual borrower characteristics that proxy for bargaining power under negotiated pricing (firm size, leverage, revenue, firm age, and relationship length), absorbing only year effects. These characteristics explain just 5% of rate variation. The second and third rows replace these controls with progressively finer fixed effects for the bank's pricing cell. At the finest level, matching the model's pricing structure (bank × borrower group × market × year), fixed effects alone explain 85.3% of all rate variation.[1] Therefore, our group pricing appears to capture significant price variation.

Another approach to measure fit is to consider the variation of interest rates within the fine cell. The median within-cell standard deviation of interest rates is 0.28 percentage points, compared to an overall cross-sectional standard deviation of 3.7 pp. If banks negotiated rates individually, borrowers within the same cell who differ in size, leverage, or relationship history would pay meaningfully different rates, and within-cell dispersion would account for a substantial share of total rate variation. Instead, within-cell dispersion is just 7.6% of the overall spread, indicating that nearly all variation in loan rates reflects which bank-group-market-year cell a borrower falls into rather than borrower-level negotiation within that cell.

Despite the good fit, two important issues could affect our exercise. First, we model pricing as a take-it-or-leave-it setting, where banks post prices and firms choose who to borrow from. In practice, lending may be subject to negotiation (Allen et al., 2019). To assess how relevant this force might be for our setting, in the last row of Table C1, we control for the individual borrower characteristics that would proxy for bargaining power. Relative to our original fit, the R^2 increases by just 0.003.

[1] This fit compares favorably to the existing literature on price imputation based on firm characteristics. For instance, Ioannidou et al. (2022) report a prediction fit of 0.85, while Crawford et al. (2018) and Degryse and Ongena (2005) find R^2 values of 0.72 and 0.67, respectively.

TABLE C1: EXPLAINED VARIANCE OF LOAN-LEVEL INTEREST RATES

The table reports the R^2 from regressing loan-level pre-tax nominal interest rates on the indicated specification. "Individual characteristics" include log assets, log debt, log revenue, firm age, and relationship length. "Coarse cell" is defined as bank×segment×region×year. "Fine cell" matches the model's pricing cell: bank×risk×loan style×segment×maturity bucket×region×year. Sample restricted to actual loans in the demand estimation sample.

	Fixed Effects	Individual Controls	R^2	N
Individual characteristics	Year	Assets, debt, revenue, firm age, rel. length	0.051	1,036,126
Coarse cell FE	Bank×Segment×Region×Year	—	0.391	1,035,931
Fine cell FE	Bank×Group×Market×Year	—	0.853	1,034,248
Fine cell FE + individual	Bank×Group×Market×Year	Assets, debt, revenue, firm age, rel. length	0.856	1,034,248
Overall SD of interest rates			3.70 pp	
Median within-cell SD			0.28 pp	
Number of fine cells			10,410	

Second, the remaining part of the price that is not perfectly predicted could come from "soft" information which may be correlated with risk. We test the quantitative importance of this concern. Following Crawford et al. (2018), we examine whether the variation in prices unexplained by our model predicts subsequent firm default. We regress standardized loan default on the same controls and the standardized residuals from the regression reported in Table C1 in the *Fine cell FE + individual*. The point estimate is 0.0039 (standard error 0.00195), with robust standard errors clustered at the bank-province-year level. Although the coefficient is statistically significant, the economic magnitude is negligible. We find that a one standard deviation increase in price residuals increases default by only 0.4% of a standard deviation, a magnitude similar to the best specifications in Crawford et al. (2018). This indicates that adverse selection unobserved to the bank is not a first order concern. Given this, and overall low default rates in our sample, we abstract from modeling adverse selection as unobserved correlated shocks between demand and default, and instead treat default as a parametric function of firm characteristics.

Besides the low economic significance of moral hazard, we show the policy (tax) has a muted direct effect on default. Table C2 shows the evolution of default before and after the policy with bank fixed effects (Column (1)) and with loan category-bank-province fixed effects (Column (2)). In both cases, the effect is small and statistically insignificant.

TABLE C2: POLICY EFFECTS ON DEFAULT

This table reports estimates of the effect of the SOLCA tax reintroduction on the probability that a loan ever defaults. Data are at the loan level. *Post* is an indicator for quarters after the tax introduction. "Category interactions" controls for the full interaction of credit purpose $\in$ {Loan, Other}, segment $\in$ {Corporate, Enterprise, SMEs, Other Commercial}, risk class $\in$ {Risky (rating $<$ *B*), Non-risky}, and maturity bucket (four bins). Standard errors clustered at the bank-quarter level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)
Post	-4.33×10^{-5}	-1.75×10^{-4}
	(4.42×10^{-4})	(4.44×10^{-4})
Constant	5.15×10^{-3}***	5.03×10^{-3}***
	(2.77×10^{-4})	(2.16×10^{-4})
Bank FE	Yes	No
Category × Bank × Province	No	Yes
Observations	757,375	756,365
R^2	0.014	0.368

Appendix C.2 Price Prediction

We assume that borrowers in a given group *g* receive offers from all banks in the region that have active lending history within that group in any year. This assumption implies that if a bank did not extend credit to a specific bin in a given year, we must impute the counterfactual offer price. We perform this imputation hierarchically, progressively relaxing the group definition: (1) we first use the average price offered by the bank in that region for the borrower's credit purpose, segment, and risk category (dropping the maturity requirement); (2) if that is unavailable, we use the average price matching credit purpose and segment only; and (3) failing that, we use the average price matching credit purpose alone.

Figure C1 presents the distributions of observed prices, predicted prices for chosen banks, and predicted prices for non-chosen banks. The figure demonstrates that the model accurately captures the non-smooth distribution of chosen prices. Moreover, it reveals a non-trivial distribution of prices for non-chosen banks, which tend to be slightly higher than the chosen ones.

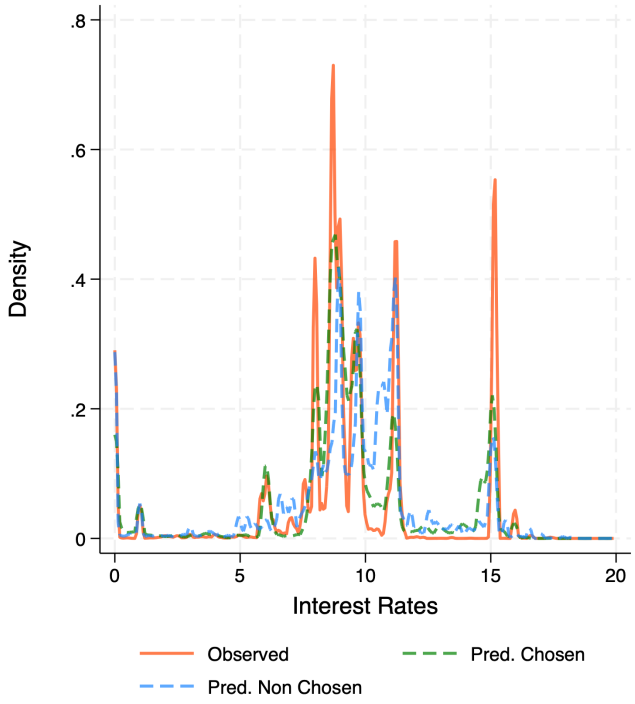


FIGURE C1: DISTRIBUTION OF PREDICTED PRICES

The figure reports the distributions of observed prices and predicted prices for borrowers' actual choices, as well as prices of non-chosen available in the choice set.

Appendix C.3 Default Prediction

Table C3 presents the estimated default prediction model, estimated on realized loans. Some results are intuitive: firms with greater assets or longer firm-bank relationships are less likely to default. Firms that have been classified as risky by the banks or as micro-firms in the commercial setting are more likely to default. Firms that are captive and have only one bank are less likely to default. The model explains around 10% of default, indicating risk beyond observable characteristics. Model fit is presented in Figure F3 in Online Appendix F.3, demonstrating a good aggregate match between realized default and predicted default.

TABLE C3: DEFAULT PREDICTION REGRESSION

This table presents the prediction default model for the estimated sample. The dependent variable is ever default dummy and the predictors include firm and contract level characteristics, as well as fixed effects. Fixed effects absorbed: bank FE, province FE, 2-digit firm sector FE.

	(1)
Firm age	0.000***
	(0.000)
Maturity (annualized)	0.005***
	(0.001)
Ln(Loan amount)	-0.003***
	(0.000)
Lagged wages	-0.001***
	(0.000)
Lagged assets	-0.002***
	(0.001)
Lagged revenue	-0.000
	(0.000)
Lagged debt	-0.000
	(0.001)
Lagged leverage	-0.000
	(0.002)
Lagged tangibility	0.002
	(0.003)
Total banks	-0.001
	(0.001)
Relationship age	-0.001**
	(0.000)
Risky	0.153***
	(0.003)
1(Micro firm)	0.017***
	(0.006)
1(Only 1 lender)	-0.005**
	(0.002)
Constant	0.071***
	(0.005)
Observations	39,354
R-squared	0.098
Bank FE	Yes
Province FE	Yes
Sector FE	Yes

Appendix C.4 Firm matching model

To incorporate the outside option into our model, we include a set of non-borrowing firms in the sample. Given the observable differences between borrowers and non-borrowers (see Table 3), we employ propensity score matching to select a comparable group of non-borrowing firms. This approach follows the strategy established in the literature to address this empirical challenge, including Adams et al. (2009) and Crawford et al. (2018). We match borrowing firms to non-borrowing firms based on observable characteristics. We then assign these matched non-borrowers to the borrowing firm's group g, endowing them with the same choice set of available banks and corresponding group-level prices, as well as the loan maturity and amount observed for the matched borrowing pair. Table C4 reports diagnostics on the quality of the match.

TABLE C4: PROPENSITY SCORE MATCHING - BIAS

The table compares the control and treatment groups before and after propensity score matching over a variety of firm-level characteristics.

	Unmatched	Mean			% Reduction	t-test	
VARIABLE	Matched	Treated	Control	% bias	in bias	t	p>t
Age - Bucket 1	U	0.17432	0.32508	-35.4		-31.50	0.000
	M	0.17432	0.1734	0.2	99.4	0.54	0.588
Debt - Bucket 1	U	0.0885	0.2409	-42.0		-41.58	0.000
	M	0.0885	0.08825	0.1	99.8	0.20	0.844
Assets - Bucket 1	U	0.08917	0.22795	-38.7		-37.85	0.000
	M	0.08917	0.08918	-0.0	100.0	-0.01	0.994
Sales - Bucket 1	U	0.07668	0.23011	-43.6		-44.37	0.000
	M	0.07668	0.07584	0.2	99.5	0.71	0.480
Wages - Bucket 1	U	0.08813	0.24507	-43.1		-42.86	0.000
	M	0.08813	0.08627	0.5	98.8	1.47	0.141
Age - Bucket 2	U	0.37793	0.36969	1.7		1.37	0.171
	M	0.37793	0.37787	0.0	99.3	0.03	0.978
Debt - Bucket 2	U	0.42857	0.4533	-5.0		-4.03	0.000
	M	0.42857	0.43142	-0.6	88.5	-1.29	0.198
Assets - Bucket 2	U	0.44012	0.45963	-3.9		-3.17	0.002
	M	0.44012	0.44109	-0.2	95.0	-0.44	0.663
Sales - Bucket 2	U	0.38715	0.46424	-15.6		-12.74	0.000
	M	0.38715	0.38827	-0.2	98.5	-0.52	0.604
Wages - Bucket 2	U	0.40019	0.48971	-18.1		-14.71	0.000
	M	0.40019	0.40187	-0.3	98.1	-0.77	0.444
Age - Bucket 3	U	0.44775	0.30522	29.7		23.21	0.000
	M	0.44775	0.44873	-0.2	99.3	-0.44	0.660
Debt - Bucket 3	U	0.48293	0.3058	36.9		28.71	0.000
	M	0.48293	0.48034	0.5	98.5	1.16	0.245
Assets - Bucket 3	U	0.47072	0.31242	32.9		25.68	0.000
	M	0.47072	0.46974	0.2	99.4	0.44	0.661
Sales - Bucket 3	U	0.53617	0.30566	48.0		37.44	0.000
	M	0.53617	0.53588	0.1	99.9	0.13	0.897
Wages - Bucket 3	U	0.51168	0.26522	52.3		40.03	0.000
	M	0.51168	0.51186	-0.0	99.9	-0.08	0.936

Appendix D Demand Estimates by Region

TABLE D1: DEMAND PARAMETERS BY REGION: DISCRETE AND CONTINUOUS ESTIMATES

The table presents the estimated parameters by region. The first column reports estimates for the discrete choice component, while the second column reports estimates for the continuous demand component. Standard errors are reported in parentheses. "×" denotes interaction terms.

Region	Variable	Discrete Coeff. (SE)	Continuous Coeff. (SE)
Azuay	Age	-0.219 (0.111)	0.194 (0.061)
	Assets	0.632 (0.798)	4.318 (0.831)
	Debt	0.941 (0.807)	1.533 (0.831)
	Capital	0.416 (0.331)	-0.584 (0.282)
	Revenue	-3.114 (0.656)	0.509 (0.590)
	Wages	3.098 (0.521)	-0.249 (0.426)
	Expenditures	2.081 (0.585)	0.247 (0.512)
	Age Relationship	0.406 (0.021)	0.155 (0.017)
	Age × Int. Rate	-0.028 (0.009)	-0.015 (0.006)
	Assets × Int. Rate	-0.375 (0.065)	0.418 (0.063)
	Debt × Int. Rate	0.040 (0.065)	-0.235 (0.063)
	Capital × Int. Rate	0.038 (0.027)	-0.032 (0.022)
	Revenue × Int. Rate	0.033 (0.053)	-0.022 (0.048)
	Wages × Int. Rate	0.140 (0.043)	-0.002 (0.035)
	Expend. × Int. Rate	0.020 (0.049)	0.002 (0.041)
	Age Rel. × Int. Rate	0.001 (0.002)	-0.010 (0.002)
	Interest Rate σ	0.327 (0.005)	
	Constant σ	0.983 (0.052)	
	Continuous σ		1.771 (0.017)
Costa	Age	-0.407 (0.088)	-0.046 (0.056)
	Assets	1.060 (0.512)	7.295 (0.492)
	Debt	-0.300 (0.520)	-1.802 (0.508)
	Capital	0.122 (0.246)	-0.724 (0.211)
	Revenue	-0.890 (0.474)	-0.399 (0.484)
	Wages	1.545 (0.282)	0.109 (0.265)
	Expenditures	0.413 (0.459)	0.086 (0.481)
	Age Relationship	0.323 (0.017)	0.206 (0.013)
	Age × Int. Rate	-0.012 (0.008)	0.006 (0.005)
	Assets × Int. Rate	-0.011 (0.040)	0.235 (0.037)
	Debt × Int. Rate	-0.032 (0.042)	-0.033 (0.038)

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TABLE D1 – continued from previous page

Region	Variable	Discrete Coeff. (SE)	Continuous Coeff. (SE)
	Capital × Int. Rate	0.023 (0.020)	-0.028 (0.017)
	Revenue × Int. Rate	0.042 (0.041)	0.029 (0.038)
	Wages × Int. Rate	0.093 (0.024)	-0.024 (0.021)
	Expend. × Int. Rate	0.000 (0.039)	0.029 (0.037)
	Age Rel. × Int. Rate	0.002 (0.002)	-0.012 (0.001)
	Interest Rate σ	0.292 (0.003)	
	Constant σ	0.993 (0.033)	
	Continuous σ		1.839 (0.014)
Guayas	Age	0.008 (0.127)	-0.024 (0.053)
	Assets	-1.633 (0.550)	9.558 (0.441)
	Debt	1.525 (0.580)	-3.443 (0.461)
	Capital	0.102 (0.283)	0.606 (0.179)
	Revenue	-3.300 (0.763)	0.465 (0.530)
	Wages	4.111 (0.364)	-0.484 (0.263)
	Expenditures	1.048 (0.664)	-0.790 (0.429)
	Age Relationship	0.272 (0.011)	0.247 (0.009)
	Age × Int. Rate	-0.033 (0.009)	0.005 (0.005)
	Assets × Int. Rate	-0.243 (0.045)	0.064 (0.035)
	Debt × Int. Rate	0.013 (0.047)	0.127 (0.037)
	Capital × Int. Rate	0.055 (0.022)	-0.110 (0.015)
	Revenue × Int. Rate	0.078 (0.059)	-0.006 (0.043)
	Wages × Int. Rate	0.021 (0.029)	0.004 (0.022)
	Expend. × Int. Rate	0.064 (0.051)	0.062 (0.035)
	Age Rel. × Int. Rate	0.003 (0.001)	-0.018 (0.001)
	Interest Rate σ	0.391 (0.005)	
	Constant σ	0.974 (0.063)	
	Continuous σ		1.826 (0.013)
Pichincha	Age	0.362 (0.161)	0.138 (0.046)
	Assets	1.994 (0.653)	10.001 (0.546)
	Debt	-3.409 (0.721)	-2.537 (0.539)
	Capital	1.313 (0.354)	-0.624 (0.174)
	Revenue	0.915 (0.624)	-0.622 (0.418)
	Wages	2.955 (0.411)	-0.692 (0.252)
	Expenditures	-0.013 (0.648)	0.189 (0.360)
	Age Relationship	0.344 (0.012)	0.126 (0.008)
	Age × Int. Rate	-0.063 (0.010)	-0.009 (0.004)
	Assets × Int. Rate	0.144 (0.062)	0.002 (0.043)

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TABLE D1 – continued from previous page

Region	Variable	Discrete Coeff. (SE)	Continuous Coeff. (SE)
	Debt × Int. Rate	0.075 (0.066)	0.024 (0.043)
	Capital × Int. Rate	-0.169 (0.031)	0.004 (0.014)
	Revenue × Int. Rate	0.098 (0.059)	0.040 (0.034)
	Wages × Int. Rate	0.357 (0.038)	0.072 (0.021)
	Expend. × Int. Rate	-0.100 (0.058)	0.025 (0.030)
	Age Rel. × Int. Rate	-0.004 (0.001)	-0.009 (0.001)
	Interest Rate σ	0.650 (0.006)	
	Constant σ	1.000 (0.076)	
	Continuous σ		1.769 (0.013)
Sierra	Age	0.158 (0.085)	0.117 (0.059)
	Assets	0.715 (0.586)	6.861 (0.587)
	Debt	0.351 (0.563)	-0.650 (0.594)
	Capital	0.096 (0.307)	0.052 (0.280)
	Revenue	-3.416 (0.810)	-1.045 (0.702)
	Wages	1.231 (0.456)	-1.772 (0.391)
	Expenditures	3.259 (0.728)	1.296 (0.628)
	Age Relationship	0.335 (0.017)	0.197 (0.015)
	Age × Int. Rate	-0.059 (0.007)	-0.019 (0.005)
	Assets × Int. Rate	-0.028 (0.047)	0.162 (0.044)
	Debt × Int. Rate	-0.001 (0.045)	-0.043 (0.044)
	Capital × Int. Rate	0.012 (0.025)	-0.089 (0.021)
	Revenue × Int. Rate	0.123 (0.062)	0.071 (0.053)
	Wages × Int. Rate	0.158 (0.036)	0.120 (0.030)
	Expend. × Int. Rate	-0.176 (0.057)	-0.057 (0.048)
	Age Rel. × Int. Rate	0.004 (0.002)	-0.006 (0.001)
	Interest Rate σ	0.347 (0.003)	
	Constant σ	0.990 (0.034)	
	Continuous σ		1.831 (0.016)

Appendix E ASOBANCA Antitrust Action

This appendix describes the antitrust activity against the Asociación de Bancos Privados del Ecuador (ASOBANCA or BA) and presents supporting evidence on lending rate patterns associated with BA membership. This institutional and empirical evidence complements Section 3.3, presenting reduced-form heterogeneity by BA commercial lending share, and Section 5.1, which reports tests of profit internalization within a candidate cartel based on BA membership against a null model of Bertrand-Nash competition.

Appendix E.1 Competition Law Background and the SCPM Action

Competition law in Ecuador is relatively new. Its guiding legislation, the *Ley Orgánica de Regulación y Control del Poder de Mercado* (LORCPM), was passed in 2011.[1] It established a dedicated competition authority, the *Superintendencia de Competencia Económica* (SCPM).

In late 2016, the SCPM pursued one of its first major actions under LORCPM. It was against the BA and all its member banks separately, for alleged unfair practices in failing to implement the national electronic payment system authorized by a 2014 financial law (Resolution SCPM-CRPI-0075-2016).

Under that 2014 law, the Central Bank of Ecuador (BCE) introduced a digital currency regulated by *Regulación Nro. 055-2014*, requiring private banks to create electronic-money accounts for users. The stated aim was to increase credit access. Similar projects, notably in Brazil and India, have demonstrated the potential of new payment systems to enhance inclusion and reduce fees (Ouyang 2022; Ghosh, Vallee, and Zeng 2024; Agarwal, Alok, Ghosh, Piskorski, and Seru 2024; Alok, Ghosh, Kulkarni, and Puri 2024). The SCPM alleged that BA coordinated member banks to boycott the system by discouraging public trust and refusing to integrate it, thus protecting their control over their deposits and payment services, a key revenue stream. BA countered that BCE-managed electronic money would risk de-dollarization and inflation.

To oppose the SCPM's resolution, BA and individual banks filed seven separate *acción de protección* lawsuits.[2] The main argument was that the SCPM lacked authority to mandate adoption of electronic money, since oversight belonged to the Central Bank, the Monetary Board, and the Banking Regulator.

1. Asociación de Bancos Privados del Ecuador "ASOBANCA" versus SCPM, Juicio No. 17203-2016-14498, brought in the Provincial Court of Justice of Pichincha on February 16, 2017.

2. Banco de la Producción S.A. PRODUBANCO versus SCPM, Juicio No. 09333-2016-00867, brought in the Multicompetent Judicial Unit, Samborondón canton, Guayas Province on February 21, 2017.

3. Banco de Loja versus SCPM, Juicio No. 17573-2016-00710, brought in the Judicial Unit for Violence Against Women and Family No. 3, Quito canton, Pichincha Province on February 24, 2017.

4. Banco Bolivariano versus SCPM, Juicio No. 09209-2016-06928 brought in the Labor Chamber of the Provincial Court of Justice of Guayas on March 6, 2017.

5. Banco Guayaquil versus SCPM, Juicio No. 09281-2016-07106 brought in the Labor Chamber of the Provincial Court of Justice of Guayas on April 3, 2017.

[1] By contrast, Colombia adopted its first competition law in 1959, and Peru in 1991.

[2] Such actions in Ecuador can be lodged in any first-instance constitutional court, even those specialized in labor, family, or domestic-violence matters.

6. Banco General Rumiñahui S.A. versus SCPM, Juicio No. 17230-2016-19522 brought in the Criminal Chamber of the Provincial Court of Justice of Pichincha on April 4, 2017.

7. Banco Pichincha C.A. versus SCPM, Juicio No. 17203-2016-14600 brought in the Criminal Chamber of the Provincial Court of Justice of Pichincha on May 18, 2017.

Only Banco Bolivariano (Juicio No. 09209-2016-06928) prevailed on appeal. On March 6, 2017, the Provincial Court of Guayas overturned the trial judge's denial and nullified the SCPM's orders, holding that the regulator had exceeded its legal remit.[3]

In 2017, the SCPM sought extraordinary review (*acción extraordinaria de protección*). On August 10, 2022 the Constitutional Court dismissed its appeal (Case No. 2463-17-EP).[4] The Court ruled narrowly that constitutional free-expression rights were implicated, justifying the lower court's acceptance of the acción de protección.

In the meantime, the BCE shut down its e-money platform, as bank adoption remained negligible. The BCE General Manager specifically cited bank opposition as the main cause. Before closing it down, the government of President Lenín Moreno tried to compromise with the banks by proposing that the BCE would no longer administer the system but that it would instead be administered by the private banks themselves.[5] Contemporary media coverage was split on the issue, but widely noted that BA had successfully opposed a major government program, that no individual member bank chose to adopt the electronic money system independently, and that the association ultimately faced no consequences. Commentators noted that these outcomes raised questions about regulatory capture or institutional weakness.

In 2022, the Banking Regulator (Superintendencia de Bancos) also voiced concerns that BA's policy proposals to the National Assembly might have anti-competitive effects.[6] A concrete example the regulator gave was how BA and its members heavily lobbied the administration of then-president Guillermo Lasso as it debated easing interest rate controls to spur credit. The Bank Regulator emphasized that any agreement among banks on setting interest rates or fees would be unacceptable and subject to antitrust action. Similarly, when BA published a "road map for financial inclusion" in 2022, proposing sector-wide strategies, regulators worried that parts of it (like standardizing certain fees or credit evaluation criteria across banks) could inadvertently limit competition if adopted collectively. The Bank Regulator's official stance was that BA should focus on education and best practices, but leave pricing and business decisions to each bank individually to preserve competitive dynamics. These episodes do not conclusively establish coordinated loan pricing in Ecuador's commercial lending markets, but illustrate BA's capacity to organize collective positions and mobilize its members on

[3] Violating Article 226 of the Constitution, which confines public agencies to their lawful scope.

[4] See http://esacc.corteconstitucional.gob.ec/storage/api/v1/10_DWL_FL/e2NhcnBldGE6J3RyYW1pdGUnLCB1dWlkOidkMDk0ZGNiZC1jOGE1LTQ2MDAtODBiZS1hYTllZDU4NDNlODkucGRmJ30= [accessed 30 Aug. 2023].

[5] See "La banca privada manejará el dinero Electrónicos." El Comercio. August 30, 2017, last accessed at https://www.elcomercio.com/actualidad/negocios/dinero-electronico-bce-banca-codigomonetario.html? in March 2025.

[6] http://esacc.corteconstitucional.gob.ec/storage/api/v1/10_DWL_FL/e2NhcnBldGE6J2VzY3JpdG8nLCB1dWlkOic5NTk2YWE0ZC04YmIxLTQ1OGMtYjg0NC00NDRlNTA3YzQ5NmMucGRmJ30= [accessed 30 Aug. 2023].

matters of shared commercial interest.

Appendix E.2 Interest Rate Dispersion by BA Membership

We can also see consistent patterns in the raw data. For example, Figure E1 illustrates how dispersed the nominal interest rates on newly granted-loans are for BA member banks (blue) and non-BA banks (orange). For each month in the sample, we compute the cross-sectional standard deviation of each bank's average lending rate. This measure indicates how tightly (or loosely) the banks' effective lending rates cluster around a common level within each group.

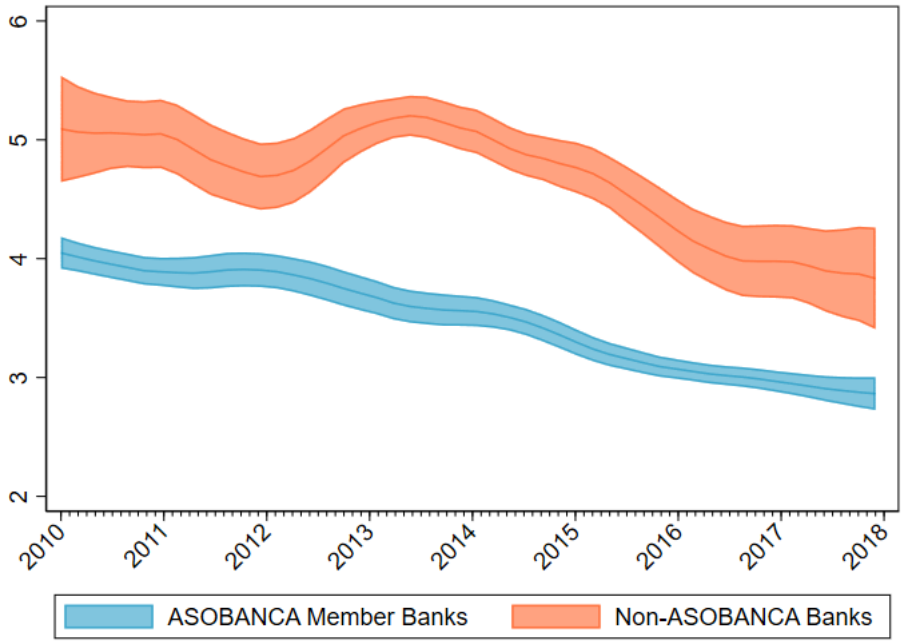


FIGURE E1: DISPERSION OF INTEREST RATES FOR ASOBANCA VERSUS NON-ASOBANCA LENDERS

For each month, the figure reports the cross-sectional standard deviation of bank-level average pre-tax nominal interest rates on newly granted commercial loans, computed separately for lenders in the Asociación de Bancos del Ecuador (ASOBANCA) (blue) and non-ASOBANCA private lenders (orange).

The figure shows two key patterns. First, the dispersion of interest rates declines over the sample period for both groups, suggesting that the range of rates across banks narrows over time. Second, throughout the entire period, BA member banks (blue) exhibit lower cross-sectional standard deviations than non-BA banks (orange), implying that BA members' rates are more similar to one another at any given point. In other words, these raw data are consistent with BA member banks' lending rates clustering more tightly than those of non-members.

This pattern provides descriptive evidence of relatively homogeneous pricing within BA member banks. It does not by itself establish coordination. BA member banks may simply have more uniform lending guidelines or customer profiles. Nonetheless, taken together with the

structural estimates in the main paper and the narrative evidence that BA has served as a vehicle for collective action among its members, these patterns are consistent with the hypothesis that BA membership is associated with coordinated conduct in lending markets.

Appendix E.3 Rate Dynamics Around BA Entry and Exit

A useful test of whether BA membership is associated with lending coordination is to examine what happens when a bank leaves or joins the association. The main such episode in our sample period is the departure of Banco Guayaquil (BG), one of Ecuador's largest private banks, from BA in May 2009, and its subsequent return around 2013.[7] Because our loan-level commercial credit data begin only in 2010, we cannot study this episode with the micro data. We therefore turn to bank-level financial statements, which are available over a longer horizon and allow us to construct implicit lending rates spanning the full episode.

We construct monthly implicit lending rates on regular commercial loans for all private banks over January 2007 to December 2017 using bank financial statements from Ecuador's Superintendencia de Bancos (*Boletín Financiero*). For each bank b, month t, and loan category c, we compute an annualized implicit lending rate as:

$$r^c_{b,t} = \frac{\text{Interest Income}^c_{b,t}}{\overline{\text{Portfolio}}^c_{b,t}} \times 12 \times 100, \tag{E1}$$

where $\text{Interest Income}^c_{b,t}$ is the monthly flow of interest income from the commercial loan portfolio and $\overline{\text{Portfolio}}^c_{b,t}$ is the average of beginning- and end-of-month gross commercial loan portfolio (before provisions). We winsorize each bank's monthly rate at its own 1st and 99th percentiles, then aggregate to quarterly frequency by taking within-quarter means. We report two group averages: an equal-weighted (EW) simple average of bank-level rates, and a portfolio-weighted (PW) ratio of total quarterly interest income to total average portfolio (annualized), which gives greater weight to larger banks.

The estimating sample comprises 21 private domestic banks. We exclude private banks with zero or negligible non-micro commercial portfolios, foreign-bank subsidiaries incorporated as private Ecuadorian banks, and Banco Pacífico.[8] We classify banks by their membership in BA, Ecuador's private banking association, which serves as a forum for lobbying and information sharing among its members.

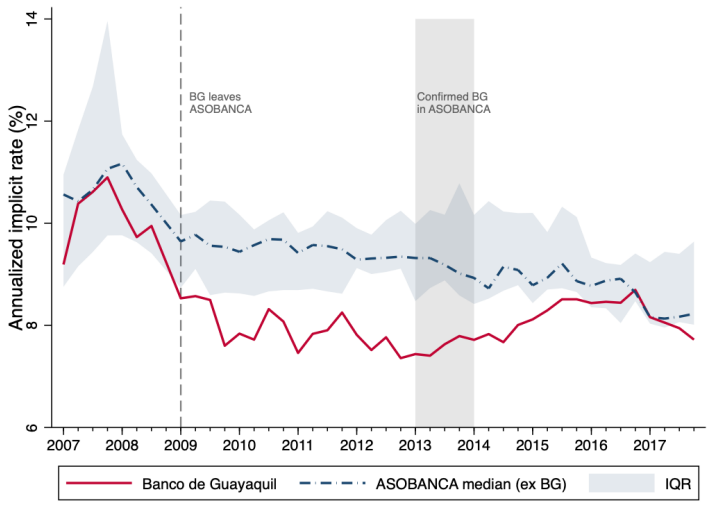
Of the 23 private, commercial-regular-loan-granting banks in Ecuador over our sample (2010-2017), 11 were members of the Asociación de Bancos Privados del Ecuador (ASOBANCA or BA) over our sample period. Of these, nine (Pichincha, Produbanco, Bolivariano, Interna-

[7] "Banco Guayaquil fuera de la ABPE," *El Universo*, May 21, 2009. Link. Per BA's own records, BG was not listed as a member in 2010. We have no membership records for 2011–2013. But BG had likely rejoined by 2013, as its president was listed on the association's board. Note that ASOBANCA is the brand name of the banking association while its acronym is ABPE.

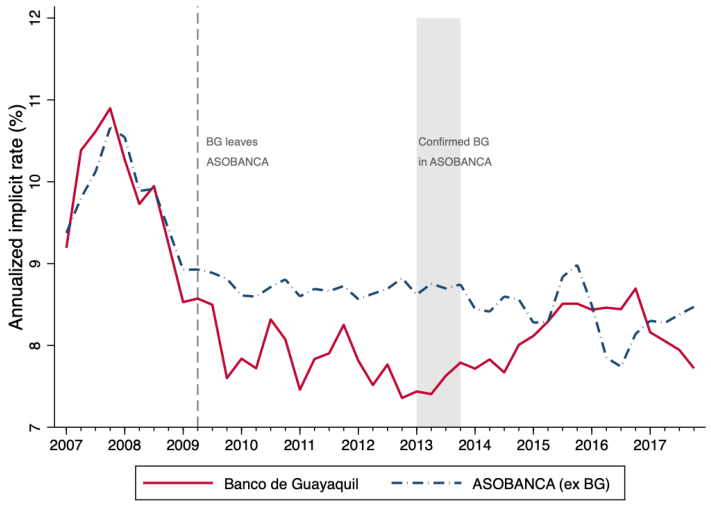
[8] Banco Pacífico (BP) was taken over by the government following a financial crisis in 2001, though it is regulated as a private bank. It was a member of BA until 2008. Since it had lower-than-average interest rates relative to other BA-member banks before 2009, including it amplifies the post-2009 difference between BG and the remaining BA members.

cional, Austro, General Rumiñahui, Machala, Loja, and Procredit) remain members throughout the sample. Promerica was a member from 2010 until it was acquired by Produbanco in 2014. One bank, Guayaquil, exits in 2009 and returns by 2013. The remaining twelve are never members within our sample.

Bank-level implicit rates. Panel (a) of Figure E2 plots Banco Guayaquil's (BG) implicit commercial lending rate (red line) against the interquartile range (IQR) of BA member rates excluding BG (blue dot-dash line). The dashed vertical line marks BG's departure from BA in mid-2009. The next time BG is listed as a member is 2013 (shaded region), when the BG president is listed on the organization's board. Prior to BG's departure, both series track closely at approximately 9.5-11%. Following BG's exit in mid-2009, BG's implicit rate drops to approximately 7.5%, while the BA member average remains flat. Throughout BG's absence, its rate falls near or below the 25th percentile of member rates. The IQR among members is relatively narrow (approximately 2 percentage points), indicating that member rates are clustered. Upon BG's return to BA around 2013, the gap narrows and the two series converge by 2014.



(a) Median and interquartile range



(b) Equal-weighted mean

FIGURE E2: IMPLICIT LENDING RATE: BANCO GUAYAQUIL VS. BA

Panel (a) reports BG's rate against the median and interquartile range of BA member rates. Panel (b) plots the quarterly equal-weighted mean implicit commercial lending rate for Banco Guayaquil (red solid) and other BA members excluding BG (blue dot-dash). The dashed vertical line marks BG's departure from BA (~May 2009); the shaded region marks the period around its confirmed return (~2013). Implicit rates are annualized: (monthly interest income/average gross commercial portfolio) × 12 × 100. Rates are winsorized at each bank's 1st and 99th percentiles before quarterly aggregation.

Panel (b) of Figure E2 plots Banco Guayaquil's (BG) implicit commercial lending rate (red line) against the equally-weighted average implicit rates of BA member rates excluding BG (blue dot-dash line). Again, a gap of roughly 1-1.5 percentage points persists throughout BG's absence from the association and the gap converges to zero after BG rejoins BA. The portfolio-weighted comparison (Figure E3) yields a similar pattern of divergence and convergence, in-

dicating that the results are not driven by small banks receiving disproportionate weight in the equal-weighted averages.

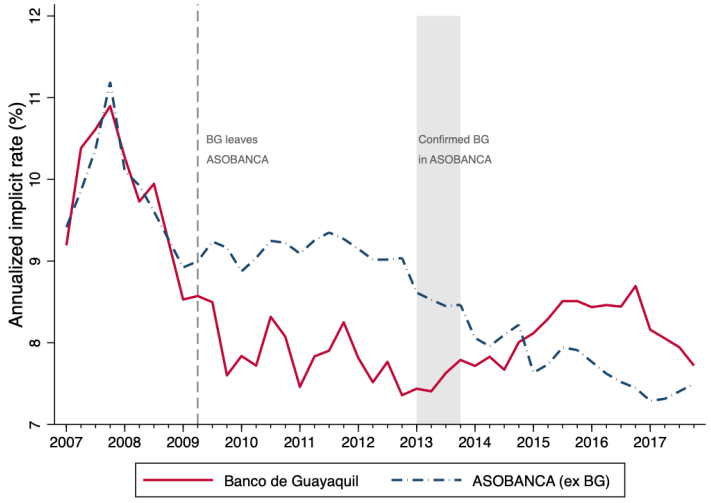


FIGURE E3: IMPLICIT LENDING RATE: BANCO GUAYAQUIL VS. BA (PORTFOLIO-WEIGHTED)

The figure reports the quarterly average implicit commercial lending rate for Banco Guayaquil (red solid) and the portfolio-weighted average for BA members excluding BG (blue dot-dash). The portfolio-weighted rate is computed as the ratio of total quarterly commercial interest income to total average commercial portfolio within each group, annualized. The dashed vertical line marks BG's departure from BA (~May 2009); the shaded region marks the period around its confirmed return (~2013).

Loan-level event study. We complement the bank-level implicit rate analysis with a loan-level event study using individual loan origination records from the Superintendencia de Bancos. The sample consists of all new Commercial Regular loans originated by BA member banks and Banco Guayaquil over 2010-2017. We winsorize each bank's rate distribution at the 1st and 99th percentiles. We estimate within loans originated by BA members:

$$r_{i,b,q} = \sum_{q \neq 2017Q3} \beta_q \cdot \mathbf{1}[\text{BG}]_b \cdot \mathbf{1}[q] + \gamma_1 \ln(\text{maturity}_i) + \gamma_2 \ln(\text{amount}_i) + \alpha_b + \delta_q + \varepsilon_{i,b,q}, \quad \text{(E2)}$$

where $r_{i,b,q}$ is the nominal interest rate on loan i from bank b in quarter q, $\mathbf{1}[\text{BG}]_b$ indicates Banco Guayaquil, α_b and δ_q are bank and quarter fixed effects, and standard errors are clustered at the bank-quarter level. The reference quarter is 2017Q3, the end of the sample well after full convergence between BG interest rates and those of other BA members.

Figure E4 plots the estimated β_q coefficients with 95% confidence intervals. We cannot test patterns around the May 2009 exit of BG from BA because our loan-level data does not extend before 2010. However, we can see that during BG's absence from the association (2010-~2012), BG charges 1-2.5 percentage points less than other BA members conditional on loan characteristics, with most quarterly estimates statistically significant at the 5% level. Following BG's return, the gap narrows gradually and becomes statistically indistinguishable from zero

by 2016-2017. These patterns are consistent with the bank-level evidence from implicit lending rates built from financial statements.

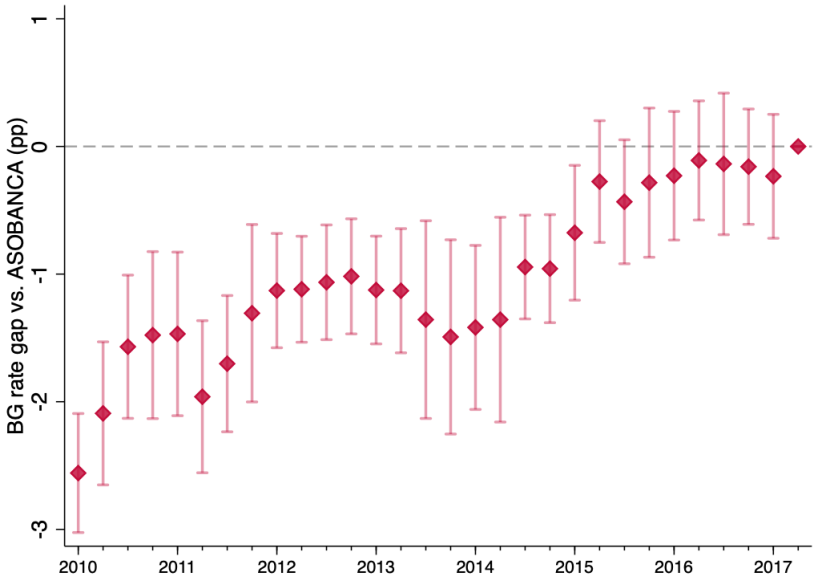


FIGURE E4: LOAN-LEVEL INTEREST RATE GAP: BANCO GUAYAQUIL VS. BA, QUARTER-BY-QUARTER

Each point reports the estimated coefficient on the interaction of a Banco Guayaquil indicator with a quarter dummy from a loan-level regression of the nominal interest rate on bank and quarter fixed effects, log maturity, and log amount, with standard errors clustered at the bank-quarter level. The reference quarter is 2017Q3 (end of sample). The sample consists of new commercial, non-micro loans originated by BA member banks and BG over 2010-2017 (N = 1,187,829). Vertical bars show 95% confidence intervals; the dashed horizontal line marks zero (no difference from BA members).

Appendix F Estimating Demand Elasticities

Define the interest-rate slopes implied by Equation 20-23:

$$b^s_{ikmt} \equiv \frac{\partial \Pi^s_{ikmt}}{\partial r_{g(i)kmt}} = \alpha^s + \alpha^{s,x}_1 X_{it} + \alpha^{s,x}_2 X_{ikmt} + \tilde{\alpha}^s_{it}, \qquad b^l_{ikmt} \equiv \frac{\partial \ln(L_{ikmt})}{\partial r_{g(i)kmt}} = \alpha^l + \alpha^{l,x}_1 X_{it} + \alpha^{l,x}_2 X_{ikmt}. \tag{F1}$$

Here α^s and α^l are the mean price coefficients recovered from the second-step IV regression (Equation 26), while $\alpha^{s,x}_1$, $\alpha^{s,x}_2$, $\alpha^{l,x}_1$, and $\alpha^{l,x}_2$ capture observable heterogeneity in price sensitivity, and $\tilde{\alpha}^s_{it}$ is the unobserved random coefficient on price in the discrete choice.

Intensive margin (loan size) elasticity. From Equation 23, the elasticity of loan size with respect to the interest rate is

$$\epsilon^{L}_{ikmt} = \frac{\partial L_{ikmt}}{\partial r_{g(i)kmt}} \frac{r_{g(i)kmt}}{L_{ikmt}} = \frac{\partial \ln(L_{ikmt})}{\partial r_{g(i)kmt}} r_{g(i)kmt} = b^{l}_{ikmt}\, r_{g(i)kmt}. \tag{F2}$$

Extensive margin (choice probability) elasticity. Let s_{ikmt} denote the (conditional) logit choice probability implied by Equation 20. The own-price derivative satisfies

$$\frac{\partial s_{ikmt}}{\partial r_{g(i)kmt}} = s_{ikmt}(1 - s_{ikmt})\, b^{s}_{ikmt}, \tag{F3}$$

and therefore the own-price elasticity of the choice probability is

$$\epsilon^{s}_{ikmt} = \frac{\partial s_{ikmt}}{\partial r_{g(i)kmt}} \frac{r_{g(i)kmt}}{s_{ikmt}} = (1 - s_{ikmt})\, b^{s}_{ikmt}\, r_{g(i)kmt}. \tag{F4}$$

Total demand elasticity. Let $Q_{ikmt} \equiv s_{ikmt} L_{ikmt}$. Then

$$\epsilon^{Q}_{ikmt} = \frac{\partial Q_{ikmt}}{\partial r_{g(i)kmt}} \frac{r_{g(i)kmt}}{Q_{ikmt}} = \epsilon^{s}_{ikmt} + \epsilon^{L}_{ikmt}. \tag{F5}$$

Cross-price elasticities (competitor $j \neq k$). Under Equation 23, L_{ikmt} depends only on $r_{g(i)kmt}$, hence

$$\epsilon^{L,j}_{ikmt} = 0. \tag{F6}$$

For choice probabilities, the cross derivative is

$$\frac{\partial s_{ikmt}}{\partial r_{g(i)jmt}} = -s_{ikmt}\, s_{ijmt}\, b^{s}_{ijmt}, \tag{F7}$$

where $b^{s}_{ijmt} \equiv \alpha^{s} + \alpha^{s,x}_{1} X_{it} + \alpha^{s,x}_{2} X_{ijmt} + \tilde{\alpha}^{s}_{it}$. Thus, the cross-price elasticity of the choice probability is

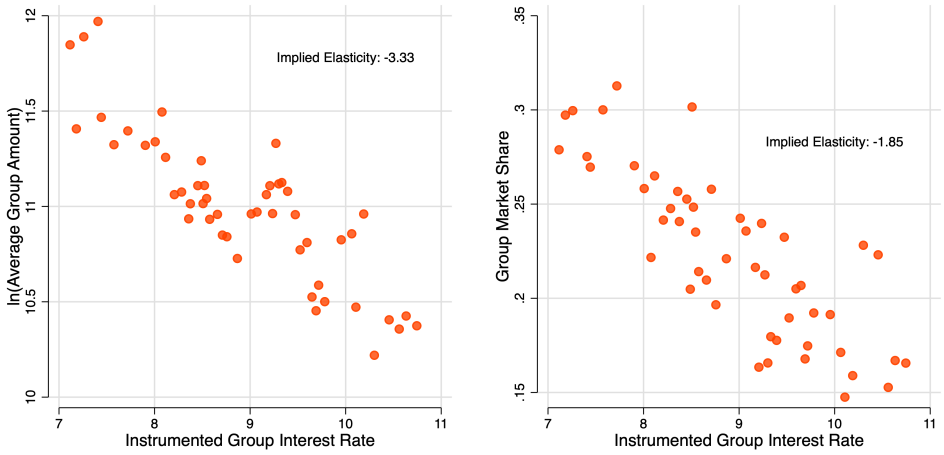
$$\epsilon^{s,j}_{ikmt} = \frac{\partial s_{ikmt}}{\partial r_{g(i)jmt}} \frac{r_{g(i)jmt}}{s_{ikmt}} = -s_{ijmt}\, b^{s}_{ijmt}\, r_{g(i)jmt}. \tag{F8}$$

Appendix F.1 Reduced-Form Elasticities

We validate the estimated structural elasticities reported in Table 7 using a reduced-form instrumental variable approach. We regress average group-level demand on instrumented average group-level interest rates, controlling for bank, province, and year fixed-effects. Instruments include delinquency rates in microcredit, housing and consumption, profitability of banking capital and interest rates in commercial credit in other regions.

We report the relationship between the instrumented interest rate and demand in Figure F1. Panel (a) presents the relationship between the instrumented interest rate and continuous de-

mand (ln(average loan value)) while panel (b) graphs the relationship with discrete-choice demand (choice probability). The corresponding implied elasticities are reported in the upper right corner of each panel. Reassuringly, the mean and median structural elasticities match the reduced-form estimates for elasticities that we calculate using an instrumental variable approach in regression form. Note that discrete choice elasticity in the reduced-form is calculated at the group-level, and thus, is less elastic than the individual-level discrete choice elasticities obtained from the structural model. For the loan use, we estimate the elasticity using average loan use, and the elasticities match well the structural values.



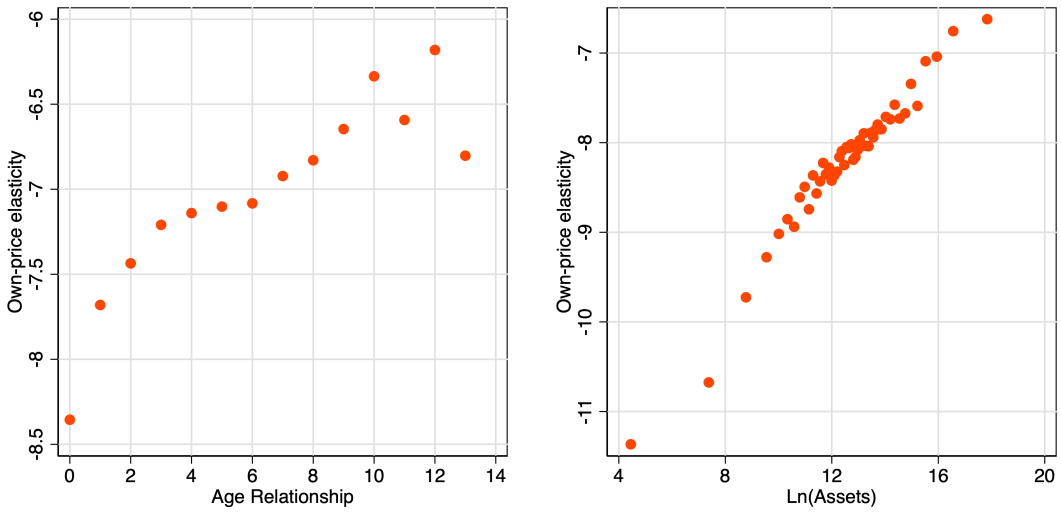
(a) Continuous (b) Discrete

FIGURE F1: REDUCED-FORM ELASTICITIES

The figure reports the reduced-form relationship between prices and demand, controlling for bank, province, and year fixed-effects. Panel A presents continuous demand (ln(average loan value)) at the borrower-group level, while Panel B presents discrete-choice demand (choice probability) at the borrower-group-level. Interest rates are instrumented using delinquency rates in microcredit, housing and consumption, profitability of banking capital, and commercial credit in other regions.

Appendix F.2 Heterogeneity by Relationship and Firm Characteristics

As an additional validation, we explore the implied heterogeneity by observed characteristics. Panel (a) of Figure F2 shows the average own-price elasticity by age of banking relationship. In line with previous literature (e.g., Allen et al. (2019)), demand becomes less elastic as the relationship between banks and borrowers deepens, possibly capturing switching costs. Panel (b) instead shows that, while demand is price elastic regardless of firm size, smaller firms are more price elastic. While this type of heterogeneity by demographics is commonly presented in retail, to our knowledge, they have not been reported for firm credit, so we cannot benchmark relative to previous literature.



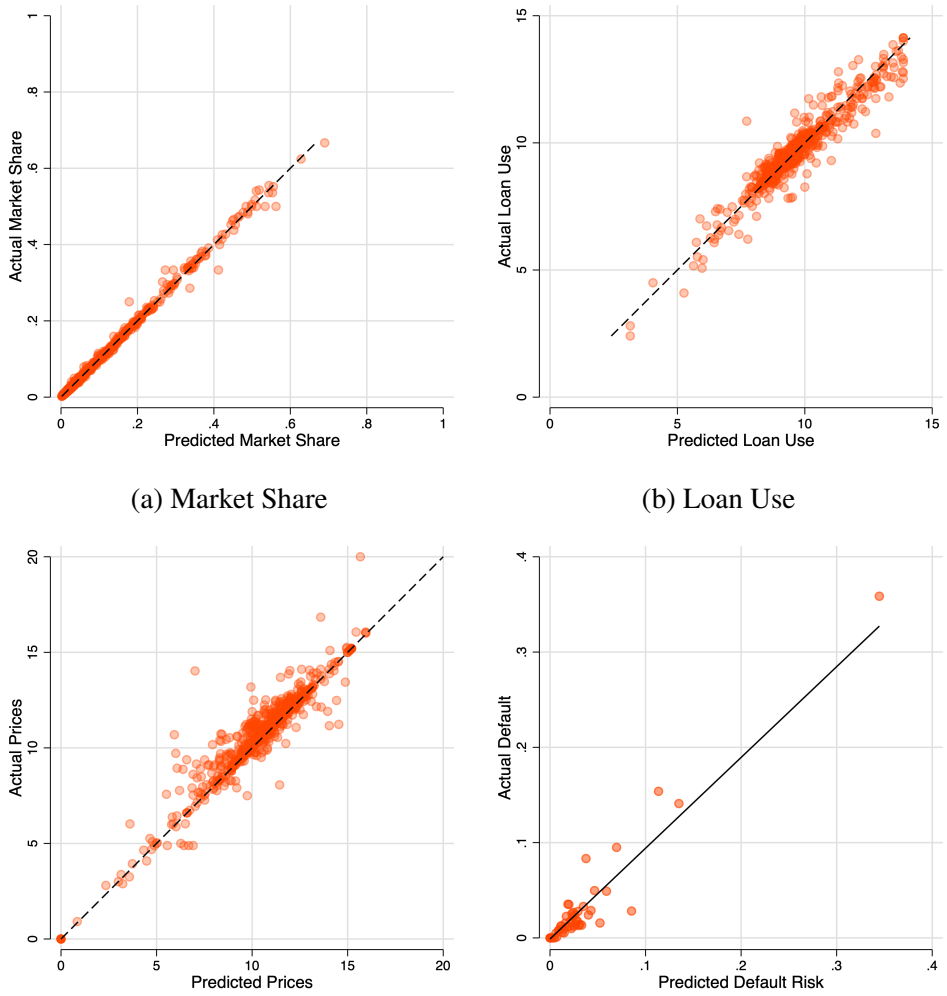
(a) Age of Relationship (b) ln(Assets)

FIGURE F2: HETEROGENEITY IN OWN-PRICE ELASTICITY BY RELATIONSHIP AND FIRM CHARACTERISTICS

The figure reports binscatter plots on the own-price elasticity by age of banking relationship and firm size (ln assets).

Appendix F.3 Model Fit

Figure F3 shows the moment fit for discrete and continuous choice, for predicted prices, and for default. The figure reports averages at the bank-region-year level. In all cases, the figures show a good fit across the full distribution.



(a) Market Share (b) Loan Use

(c) Interest Rates (d) Market Share

FIGURE F3: BANK-REGION-YEAR FIT

The figure reports binscatter plots at the bank-region-year level for observed market shares vs predicted market shares (panel a), observed average observed loan use vs. average predicted loan use (panel b), observed average prices vs predicted average prices (panel c), and observed default vs predicted default (panel d)

Appendix F.4 Heterogeneity of Pass-through by Mode of Conduct

In Table F1, we report moments of the distributions of the simulated pass-throughs under Bertrand-Nash and under $\kappa = 1$ for the alleged BA cartel. Pass-throughs are simulated as a common tax of 0.1% and calculated as the difference in pre-tax and post-tax Nash equilibrium pricing, conditional on a conduct assumption. The borrower-bank-level pass-throughs are averaged at the borrower level by using post-tax, individual-level market shares, then are averaged across borrowers at the borrower-group level. Note that the Bertrand-Nash distribution spans both sides of one (median 0.996, 75th percentile 1.008), confirming that the estimated demand system is flexible enough to generate pass-through above and below one without re-

quiring profit internalization. Incomplete pass-through is therefore not mechanically imposed by restrictive demand curvature (Miravete et al. (2023)).

TABLE F1: DISTRIBUTION OF PASS-THROUGH BY CONDUCT STATUS

The table reports pass-throughs obtained at the borrower-group-year as a simple average across borrowers. The borrower-level pass-through is calculated as the weighted average across banks in the choice set using post-tax market shares as weights. Sample restricted to years < 2015.

	Conduct	
	Bertrand-Nash	Alleged BA Cartel
Mean	0.852	0.770
P10	0.136	0.077
P25	0.548	0.407
P50	0.996	0.949
P75	1.008	1.002
P90	1.046	1.037

Appendix G Testing the Marginal Cost Assumptions

We assume that effective marginal costs used for pricing are common at the borrower-group-level for each bank-market-year, created by a combination of marginal costs common at the group-level and idiosyncratic borrower-level costs. Namely:

$$\widetilde{\mathbf{mc}}_{gmt}(\Upsilon_{gmt}) = \Big[\sum_{i\in g} \Upsilon_{gmt} \circ \mathbf{\Delta}^{d}_{imt}\Big]^{-1} \sum_{i\in g} [\Upsilon_{gmt} \circ \mathbf{\Delta}_{imt}]\, \mathbf{mc}_{imt}. \tag{G1}$$

for $\mathbf{mc}_{imt} = \bar{mc}(L_{g(i)mt}, w_{g(i)mt}) + \omega_{imt}$, where $\bar{mc}(L_{g(i)mt}, w_{g(i)mt})$ depends on total loans offered at the group level ($L_{g(i)mt} = \sum_{i\in g} L_{imt}$) and group-level cost shifters $w_{g(i)mt}$ (such as average assets or average firm age), and ω_{imt} is the idiosyncratic cost shock.

Notice the generality of our marginal cost assumption, as the model allows for considerable heterogeneity. First, we allow marginal cost to be borrower-group specific. For example, some borrower groups may be easier to monitor so that the bank will have a lower marginal cost of lending to them. This also allows for different levels of loan output at the borrower level to affect marginal costs, allowing for (dis)economies of scale. Second, we allow the marginal cost to be bank dependent, capturing differences in efficiency across banks. Third, we allow for differences across markets, permitting geographical dispersion such as that related to the density of the bank's local branches. This would control for factors such as bank specialization in lending to specific sectors. Fourth, the total cost at the group level depends on borrower-specific differences and the pool of borrowers within a group, allowing for costs to depend on this additional level of selection.

Despite this flexibility, there is a key empirical limitation of our framework. Namely, we do not have enough instruments to separately identify conduct and economies of scale (Dearing et al. (2024)). As such, we recover marginal costs at the group-bank-market-year level for each potential mode of conduct and ex-post test whether the estimated marginal costs are consistent with constant (or non-constant) marginal costs. We first derive our testing framework.

For expositional convenience, let $\mathbf{A}_{imt}(\boldsymbol{\Upsilon}_{mt}) \equiv \boldsymbol{\Upsilon}_{mt} \circ \boldsymbol{\Delta}^{d}_{imt}$, $\mathbf{A}_{gmt}(\boldsymbol{\Upsilon}_{mt}) \equiv \sum_{i \in g} \mathbf{A}_{imt}(\boldsymbol{\Upsilon}_{mt})$, $\mathbf{B}_{imt}(\boldsymbol{\Upsilon}_{mt}) \equiv \boldsymbol{\Upsilon}_{mt} \circ \boldsymbol{\Delta}_{imt}$, and $\mathbf{B}_{gmt}(\boldsymbol{\Upsilon}_{mt}) \equiv \sum_{i \in g} \mathbf{B}_{imt}(\boldsymbol{\Upsilon}_{mt})$. Plugging in mc_{imt}, we obtain borrower-group costs as

$$\widetilde{\mathbf{mc}}_{gmt}(\boldsymbol{\Upsilon}_{gmt}) = \mathbf{A}_{gmt}(\boldsymbol{\Upsilon}_{gmt})^{-1}\mathbf{B}_{gmt}(\boldsymbol{\Upsilon}_{gmt})(\overline{mc}(L_{gmt}, w_{gmt})\,\mathbb{1}) + \boldsymbol{\omega}^{\Upsilon}_{gmt}. \tag{G2}$$

where $\boldsymbol{\omega}^{\Upsilon}_{gmt} \equiv \mathbf{A}_{gmt}(\boldsymbol{\Upsilon}_{gmt})^{-1} \sum_{i \in g} \mathbf{B}_{imt}(\boldsymbol{\Upsilon}_{gmt})\,\boldsymbol{\omega}_{imt}$. We parametrize the borrower-group common function as

$$\bar{mc}(L_{gmt}, w_{gmt}) = \beta_l L_{gmt} + \beta_w w_{gmt}, \tag{G3}$$

where β_l captures the scale parameter. For $\beta_l > 0$ ($\beta_l < 0$) we obtain increasing (decreasing) marginal costs, while if $\beta_l = 0$, we have constant marginal cost. The scale parameter is relevant for our context as increasing marginal costs imply more incomplete pass-throughs (Weyl and Fabinger (2013)). Thus, our simulations would be biased if true marginal costs are increasing in the quantity demanded by the borrower while we assume they are constant. Moreover, the results of conduct testing would also be biased (Dearing et al. (2024); Duarte et al. (2025)).

Default risk implies that $A_{gmt}(\Upsilon_{mt})^{-1}B_{gmt}(\Upsilon_{mt})$ is close to, but not identical to, the identity matrix. A simple instrumental variables regression of borrower-group marginal costs on total loan sizes would therefore be biased, likely upward, because the denominator elements are a fraction of the corresponding numerator elements. This amplification bias would work against finding constant marginal costs. Moreover, except under Bertrand-Nash, the off-diagonal elements of the internalization matrix are close to, but not exactly, zero. Therefore, we must also incorporate loan volume from other banks.

For simplicity, we estimate the empirical model

$$\widehat{\widetilde{mc}}^{\Upsilon}_{gkmt} = \beta^{\Upsilon}_{l} L_{gkmt} + \beta^{\Upsilon}_{l'} L_{gk'mt} + \beta^{\Upsilon}_{w} w_{gkmt} + \alpha_k + \omega^{\Upsilon}_{gkmt}, \tag{G4}$$

where $\widehat{\widetilde{mc}}^{\Upsilon}_{gkmt}$ is the implied marginal costs for borrower-group g, bank k, market m, and year t under the conduct assumption Υ. $L_{gk'mt}$ is total lending by other banks to the same borrower-group-market-year. We instrument L_{gkmt} with a post-SOLCA indicator ($Post_t$), its interaction with a public-bank indicator ($Post_t \times Public_k$), and, for cartel models, its interaction with a cartel-membership indicator ($Post_t \times Cartel^{\Upsilon}_k$). Standard errors are clustered at the bank level.

TABLE G1: TEST OF MARGINAL COST ASSUMPTIONS

This table reports the coefficient $\widehat{\beta_l}$ from the instrumental variable model in Equation G4, along with the t-statistic, observations, and first-stage diagnostics, under four alternative conduct models: Bertrand-Nash, and the alleged BA cartel, Top-4 cartel, and Industry-Wide cartel. Data are at the borrower-group × bank-year level for 2010–2017, for realized loans. Marginal costs are from Equation G2, a weighted average of borrower × bank-year marginal costs. All models include bank fixed effects and borrower-group-bank-year averages of firm age, relationship age, log assets, log debt, log revenue, log capital, log expenditures, and log wages. Models other than Bertrand-Nash control for total amount borrowed from other banks in the region-year by the same borrower group. Amounts are in millions of 2010 USD. Robust standard errors are clustered at the bank level.

	(1) Bertrand–Nash	(2) BA	(3) Top-4	(4) Industry-Wide
Coefficient ($\widehat{\beta_l}$)	1.912	1.177	1.691	11.522
T-statistic	(0.99)	(1.18)	(1.25)	(1.52)
Observations	5,413	5,243	5,315	5,280
Weak ID (F-stat)	7.456	42.242	39.020	2.592
Under ID (LM stat)	3.163	6.678	2.834	2.249
Over ID (Hansen J)	0.088	0.645	0.824	0.000

Table G1 reports evidence consistent with constant returns to scale at the borrower-group level. We fail to reject the null hypothesis of constant marginal costs under all conduct models. Turning to instrument diagnostics, we reject underidentification, indicating that the instruments have explanatory power for loan volume. The instruments are strong for the alleged BA cartel and alleged Top-4 cartel, though they are relatively weak for Bertrand-Nash and weak for the alleged Industry-Wide cartel. The Hansen J overidentification test fails to reject the null of instrument validity, indicating that the instruments do not produce inconsistent estimates.[1] Except for the Industry-Wide cartel, all point estimates have a similar magnitude across conduct models. These estimates are likely biased upward because the simplified linear specification in Equation G4 does not incorporate the default-risk adjustment in the structural cost equation.

Even if we act conservatively and take the point estimates as statistically significant, the implied curvature is economically small. Under the alleged BA cartel, moving from the 25th to the 75th percentile of borrower-group loan volume (20,000 to 468,069 USD) increases marginal cost by 11% relative to average cost. Internet Appendix Figure B5 shows no statistically significant change in aggregate credit following the tax, implying that the tax-induced quantity shift is far smaller than the IQR and the resulting bias in implied marginal costs would be a small fraction of this 11% upper bound. Given the lack of economic and statistical significance, we do not incorporate the joint testing of economies of scale and conduct proposed by Duarte et al. (2025).

As a further robustness check, in the spirit of Bresnahan (1987), we examine the stability of the implied marginal costs over time under each candidate conduct model. Figure G1 reports the evolution of average estimated marginal costs over time under our candidate conduct models: Bertrand-Nash, alleged Top 4 Cartel, alleged banking association (BA) cartel, and alleged Industry-Wide (All) Cartel. Under the alleged BA cartel, marginal costs are broadly stable around the introduction of the tax, consistent with the maintained assumption that the SOLCA tax did not shift real lending costs. For Bertrand-Nash and the alleged Top-4 cartel, we see a slight increase in 2015. Under the Bresnahan diagnostic, this is what we would expect if the true model involves profit internalization but the econometrician forces Bertrand-Nash: the model cannot explain incomplete pass-through without pushing up implied costs. For the alleged Industry-Wide (All) cartel, implied costs are volatile throughout the sample, consistent with a model that overstates markups far beyond what the data support, forcing the cost inversion to absorb period-to-period variation in prices and demand composition.

The figure is also informative about levels. The Bertrand-Nash model yields significantly higher marginal cost levels throughout the period, diverging from observable proxies like deposit rates (Table 3), whereas the alleged BA cartel yields cost estimates that align more closely with those external benchmarks.

[1] This test cannot detect endogeneity common to all instruments.

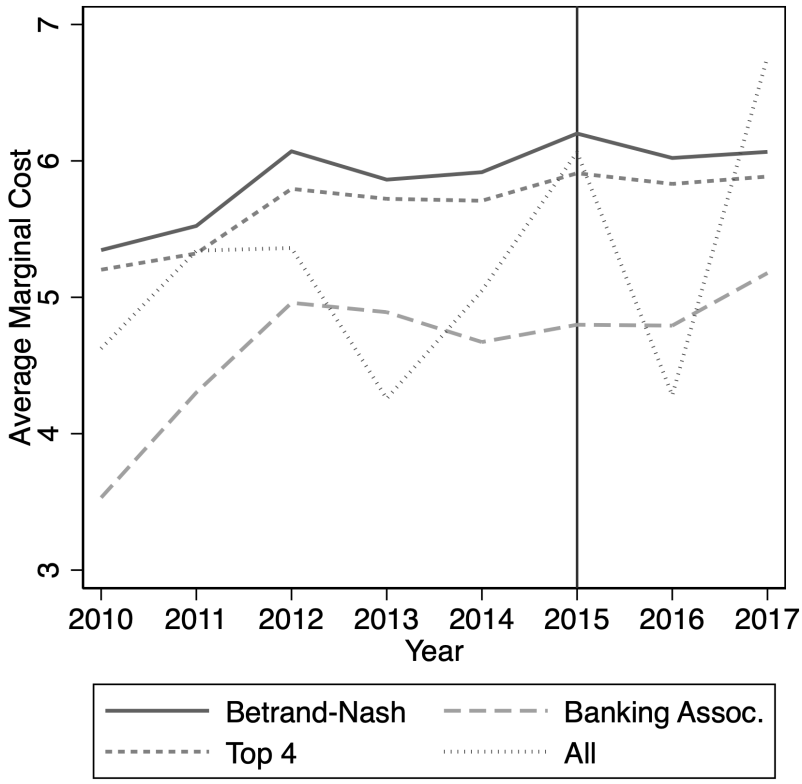


FIGURE G1: DYNAMIC ANALYSIS OF MARGINAL COSTS BY CONDUCT MODEL

The figure reports default-adjusted average marginal costs around the introduction of the SOLCA tax (represented by the solid vertical line at 2015). Marginal costs are reported separately by conduct model: Bertrand-Nash competition (solid), the benchmark; an alleged cartel of the members of the BA banking association (dashed); a cartel comprising all lenders (dot); and a cartel of the top four largest national banks (short-dashed).

Appendix H Welfare Effects of Taxation

Welfare Calculations

We compute welfare for each conduct model (Bertrand-Nash and each alleged cartel at each κ) by comparing the equilibrium without the tax (scenario 0) and with the tax (scenario 1); differences between conduct models, reported in Table H1, are differences of these changes. For each scenario $j \in \{0, 1\}$, the equilibrium vector of (pre-tax) interest rates is $\{r^j_{g(i)kmt}\}_{i,k,m,t}$, obtained from the supply model under the relevant conduct assumption. Borrowers face a tax-inclusive interest-rate wedge

$$\tilde{r}^j_{ikmt} \equiv r^j_{g(i)kmt} + \tau^j_{ikmt},$$

and all demand objects below (choice probabilities and loan size) are evaluated at $\tilde{r}^j_{ikmt}$.

Discrete (extensive-margin) borrower surplus. Let $\tilde{\Pi}^{s,j}_{ikmt}$ denote the deterministic component of the discrete indirect profit in Equation 20 evaluated at $\tilde{r}^{j}_{ikmt}$ (i.e., excluding the i.i.d. Type-I Extreme Value taste shock ε^{s}_{ikmt}):

$$\tilde{\Pi}^{s,j}_{ikmt} \equiv \tilde{\delta}^{s}_{g(i)kmt} + b^{s}_{ikmt}\,\tilde{r}^{j}_{ikmt} + \beta^{s}_{1} X_{it} + \beta^{s}_{2} X_{ikmt} + \psi^{s}_{it}, \qquad b^{s}_{ikmt} \equiv \frac{\partial \Pi^{s}_{ikmt}}{\partial r_{g(i)kmt}},$$

where b^{s}_{ikmt} is defined in Internet Appendix F. Conditional on a draw of the random coefficients and random effects (e.g., $(\tilde{\alpha}^{s}_{i}, \psi^{s}_{it})$), the logit inclusive value is

$$IV^{j}_{it} = \log\left(1 + \sum_{k \in K_{g(i)mt}} \exp\left(\tilde{\Pi}^{s,j}_{ikmt}\right)\right),$$

where the "1" corresponds to the outside option normalized to zero.

To express surplus in interest-rate units, we use the compensating-variation formula. Under the maintained approximation that the marginal disutility of the tax-inclusive rate is common across the alternatives within (i,t) for each draw,[1] the change in discrete surplus for borrower i is

$$\Delta CS^{\text{disc}}_{imt} = \mathbb{E}\left[-\frac{IV^{1}_{it} - IV^{0}_{it}}{\bar{b}^{s}_{it}}\right],$$

where the expectation is taken over the simulated draws used for the mixed-logit integration.

Continuous (intensive-margin) borrower surplus. From Equation 23, conditional loan size under scenario j is

$$\ln L^{j}_{ikmt} = \tilde{\delta}^{l}_{g(i)kmt} + b^{l}_{ikmt}\,\tilde{r}^{j}_{ikmt} + \beta^{l}_{1} X_{it} + \beta^{l}_{2} X_{ikmt} + \varepsilon^{l}_{ikmt}, \qquad b^{l}_{ikmt} \equiv \frac{\partial \ln L_{ikmt}}{\partial r_{g(i)kmt}} < 0.$$

This implies $L^{j}_{ikmt} = \exp(\cdot)$ and $\frac{\partial L_{ikmt}}{\partial r_{g(i)kmt}} = b^{l}_{ikmt} L_{ikmt}$. The Marshallian surplus associated with the intensive margin (in interest-rate units) is

$$CS^{\text{cont},j}_{ikmt} = -\frac{L^{j}_{ikmt}}{b^{l}_{ikmt}}.$$

Let s^{j}_{ikmt} denote the (mixed) discrete choice probability under scenario j and define expected relationship-level demand

$$Q^{j}_{ikmt} \equiv s^{j}_{ikmt}\, L^{j}_{ikmt}.$$

Then the change in expected continuous surplus for borrower i is

$$\Delta CS^{\text{cont}}_{imt} = \sum_{k \in K_{g(i)mt}} \left[s^{1}_{ikmt}\, CS^{\text{cont},1}_{ikmt} - s^{0}_{ikmt}\, CS^{\text{cont},0}_{ikmt} \right].$$

[1] In practice, when b^{s}_{ikmt} varies across k through X_{ikmt}, we implement the formula using the draw-specific average $\bar{b}^{s}_{it} \equiv |K_{g(i)mt}|^{-1} \sum_{k \in K_{g(i)mt}} b^{s}_{ikmt}$ (results are similar using the chosen-k value).

Total borrower surplus. We define borrower surplus as the sum of the extensive- and intensive-margin components:

$$\Delta CS_{imt} = \Delta CS^{\text{disc}}_{imt} + \Delta CS^{\text{cont}}_{imt}$$

Bank (producer) surplus. For scenario j, expected profits from lending relationship (i, k, m, t) are

$$\pi^{j}_{ikmt} = \left[(1 - d_{ikmt})\, r^{j}_{g(i)kmt} - mc_{ikmt}\right] Q^{j}_{ikmt},$$

where d_{ikmt} is the predicted default probability at origination and mc_{ikmt} denotes the (model-consistent) marginal cost used in the counterfactual.[2] Aggregating,

$$PS^{j}_{imt} = \sum_{k \in K_{g(i)mt}} \pi^{j}_{ikmt}, \qquad \Delta PS_{imt} \equiv PS^{1}_{imt} - PS^{0}_{imt},$$

summing over all banks in i's choice set.

Tax revenue. Tax revenue in scenario j is the tax wedge times expected loan volume:

$$TR^{j}_{imt} = \sum_{k \in K_{g(i)mt}} \tau^{j}_{ikmt}\, Q^{j}_{ikmt}, \qquad \Delta TR_{imt} \equiv TR^{1}_{imt} - TR^{0}_{imt}.$$

Incidence and excess burden. We report incidence shares (per dollar of revenue) for each firm-year i from its choice set totals, defined above, and Table 12 reports their means across firm-years.

$$\mathcal{I}_{imt} \equiv \frac{\Delta CS_{imt}}{\Delta PS_{imt}}, \qquad \mathcal{I}^{\text{Borrowers}}_{imt} \equiv -\frac{\Delta CS_{imt}}{\Delta TR_{imt}}, \qquad \mathcal{I}^{\text{Banks}}_{imt} \equiv -\frac{\Delta PS_{imt}}{\Delta TR_{imt}}.$$

We report the gross excess burden (deadweight loss) per dollar of revenue as

$$\mathcal{EB}_{imt} \equiv \frac{-\left(\Delta CS_{imt} + \Delta PS_{imt} + \Delta TR_{imt}\right)}{\Delta TR_{imt}} = -\frac{\Delta CS_{imt} + \Delta PS_{imt}}{\Delta TR_{imt}} - 1.$$

The net excess burden replaces ΔPS_{imt} with the bank-surplus change evaluated at the Bertrand-Nash marginal-cost schedule, $\Delta PS^{\text{net}}_{imt} = \sum_{k \in K_{g(i)mt}} [((1 - d_{ikmt})\, r^{1}_{g(i)kmt} - mc^{\text{BN}}_{ikmt}) Q^{1}_{ikmt} - ((1 - d_{ikmt})\, r^{0}_{g(i)kmt} - mc^{\text{BN}}_{ikmt}) Q^{0}_{ikmt}]$, where mc^{BN}_{ikmt} is the marginal cost recovered under Bertrand-Nash. The difference $\Delta PS_{imt} - \Delta PS^{\text{net}}_{imt} = \sum_{k \in K_{g(i)mt}} (mc^{\text{BN}}_{ikmt} - mc_{ikmt})(Q^{1}_{ikmt} - Q^{0}_{ikmt})$ is the supra-competitive margin forgone on the lost lending; the efficiency-component row of Panel A reports this term for the chosen bank.

[2] In our baseline tax counterfactual, the SOLCA tax is a wedge borne by borrowers and remitted to the government, so it enters demand through $\tilde{r}^{j}_{ikmt}$ but does not directly shift mc_{ikmt}.

Bootstrap Standard Errors

The standard errors in Table 12 and the confidence band in Figure 5 come from a parametric bootstrap over the demand estimates, which we propagate through every subsequent step of the welfare exercise. In each draw b, we perturb the three blocks of demand parameters region by region. First, we draw the nonlinear parameters θ (the random-coefficient dispersions and demographic interactions) from $N(\hat{\theta}, \hat{V}_\theta)$, where $\hat{V}_\theta$ is the θ block of the inverse of the numerical Hessian of the joint likelihood, including the fixed effects. Second, we draw the two price coefficients (on the bank-choice and loan-size margins) with replacement from 500 bank-clustered bootstrap replications of the instrumental-variables regression that recovers them, applying the same draw in every region. Third, we adjust the fixed effects to the drawn θ through the Hessian rather than re-estimating them.

Given the drawn parameters, we recover marginal costs under each conduct model, re-solve the pre- and post-tax equilibria, and recompute borrower surplus, bank surplus, and revenue for each borrower, applying the same trimming and aggregation as for the point estimates. Each draw uses the same parameters for every conduct model, so differences across columns are computed within draw. We retain draws in which both price coefficients are negative. The bootstrap holds fixed the estimation sample, the predicted default probabilities and offered prices, the conduct model within each column, and the constant-marginal-cost assumption.

Reported standard errors are the standard deviation of each statistic across the 500 retained draws. The 80% bias-corrected intervals in brackets are the 10th and 90th percentiles of the draws, shifted so that the bootstrap median coincides with the point estimate. The Figure 5 band is built the same way from the 5th and 95th percentiles of the revenue premium across 200 draws, computed at eight tax rates.

Additional Welfare Results

Table H1 reports, for each conduct specification, the difference from Bertrand-Nash in each outcome of Table 12, computed within draw, with its standard error and 80% bias-corrected interval. We use 80% intervals because the pass-through mechanism predicts the sign of most differences, so the relevant test is one-sided, and an 80% interval that excludes zero on the predicted side corresponds to a one-sided test at the 10% level.

TABLE H1: DIFFERENCES FROM BERTRAND-NASH ACROSS CONDUCT SPECIFICATIONS

This table reports differences from Bertrand-Nash in the welfare effects of a uniform loan tax of $\tau = 0.5\%$. Panel A reports differences between medians of firm-year outcomes, while Panel B reports differences between means of incidence and welfare ratios. For gross- and net-excess-burden rows, a positive difference denotes a larger welfare loss per dollar of revenue under the column conduct specification. The sample stacks five regions, restricts attention to years before 2015 and realized lending relationships, and trims each outcome at its 10th and 90th percentiles within conduct specification before calculating medians and means. Parentheses report parametric bootstrap standard errors, calculated as the standard deviation of the within-draw conduct-minus-Bertrand-Nash difference across 500 draws of demand parameters. Brackets report 80% bias-corrected bootstrap intervals.

	Top4 κ=0.2	Top4 κ=1	BA κ=0.2	BA κ=1
Panel A: Medians of Δ Outcomes				
Δ Borrower surplus (CS)	5.41	181.70	-9.35	442.04
	(132.42)	(187.30)	(145.14)	(355.67)
	[-55, 39]	[96, 333]	[-65, 53]	[152, 1,179]
Δ Bank surplus (PS)	-124.45	-705.77	-313.20	-2,253.81
	(132.04)	(285.86)	(118.74)	(563.77)
	[-151, 54]	[-932, -229]	[-359, -174]	[-2,977, -1,536]
Δ Tax revenue	11.56	150.71	15.44	330.78
	(15.25)	(19.44)	(15.98)	(20.57)
	[5, 17]	[141, 160]	[6, 22]	[312, 352]
Δ PS (efficiency component)	-2.42	-16.38	-42.02	-317.43
	(0.71)	(5.18)	(13.85)	(127.09)
	[-3, -1]	[-24, -11]	[-52, -16]	[-480, -144]
Δ Bank surplus (net)	-12.24	-0.47	-10.98	83.93
	(119.34)	(183.37)	(125.76)	(455.38)
	[-18, 110]	[-31, 138]	[-31, 44]	[-411, 206]
Panel B: Means of Incidence and DWL				
Incidence (ΔCS / ΔPS)	-0.02	-0.12	-0.05	-0.32
	(0.03)	(0.18)	(0.07)	(0.38)
	[-0.05, -0.00]	[-0.32, -0.03]	[-0.13, -0.01]	[-0.76, -0.12]
Excess Burden	0.02	0.09	0.05	0.28
	(0.03)	(0.05)	(0.03)	(0.09)
	[0.01, 0.02]	[0.03, 0.13]	[0.04, 0.08]	[0.16, 0.37]
Net Excess Burden	0.00	-0.02	0.01	-0.10
	(0.03)	(0.04)	(0.03)	(0.08)
	[-0.00, 0.01]	[-0.04, 0.00]	[0.00, 0.04]	[-0.14, -0.02]
Δ CS / Δ Revenue	-0.00	-0.02	-0.00	-0.04
	(0.02)	(0.02)	(0.01)	(0.04)
	[-0.01, 0.00]	[-0.03, -0.00]	[-0.01, 0.01]	[-0.12, -0.03]

Appendix I Additional Conduct Testing

TABLE I1: WITHIN-CARTEL TESTS: PARTIAL AGAINST FULL INTERNALIZATION

This table tests partial internalization ($\kappa < 1$, null) against full internalization ($\kappa = 1$, alternative), holding cartel membership fixed. A positive T^{RV} favors joint maximization over the null. Column 4 reports the effective F, Column 5 the critical value it must exceed for best-case power to reach 95 percent, and Column 6 the degeneracy measure ρ^2 of Duarte et al. (2024), which together with the number of instruments indexes the critical value. Worst-case size is bounded at 7.5 percent at every effective F in every comparison, so the corresponding size critical value is zero throughout and is not reported.

Panel A. Banking Association Cartel					
Null (κ)	Alternative (κ)	T^{RV}	Effective F	Critical Value	ρ^2
(1)	(2)	(3)	(4)	(5)	(6)
0.2	1.0	0.64	3.32	5.9	0.125
0.4	1.0	1.23	3.66	6.1	0.105
0.6	1.0	1.73	4.25	6.2	0.094
0.8	1.0	-0.74	4.93	5.7	0.139
Panel B. Top 4 Cartel					
Null (κ)	Alternative (κ)	T^{RV}	Effective F	Critical Value	ρ^2
(1)	(2)	(3)	(4)	(5)	(6)
0.2	1.0	-0.18	4.45	7.6	0.007
0.4	1.0	-0.02	4.30	7.6	0.006
0.6	1.0	0.35	4.07	7.9	0.001
0.8	1.0	-0.24	4.40	7.4	0.013
Panel C. Industry-Wide Cartel					
Null (κ)	Alternative (κ)	T^{RV}	Effective F	Critical Value	ρ^2
(1)	(2)	(3)	(4)	(5)	(6)
0.2	1.0	-0.93	4.85	6.1	0.211
0.4	1.0	0.07	6.58	6.8	0.143
0.6	1.0	0.83	8.62	8.1	0.049
0.8	1.0	-3.16	7.11	5.9	0.234

TABLE I2: ACROSS-CARTEL TESTS AT MATCHED INTERNALIZATION

This table compares cartel membership definitions holding the internalization parameter κ fixed. A positive T^{RV} favors the alternative (Column 2) over the null (Column 1). Column 4 reports the effective F, Column 5 the critical value it must exceed for best-case power to reach 95 percent, and Column 6 the degeneracy measure ρ^2 of Duarte et al. (2024), which together with the number of instruments indexes the critical value. Worst-case size is bounded at 7.5 percent at every effective F in every comparison, so the corresponding size critical value is zero throughout and is not reported.

Panel A. Top 4 vs. Banking Association Cartel

Null (κ) (1)	Alternative (κ) (2)	T^{RV} (3)	Effective F (4)	Critical Value (5)	ρ^2 (6)
0.2	0.2	-3.03	8.86	6.4	0.019
0.4	0.4	-2.93	8.37	6.1	0.033
0.6	0.6	-2.87	7.65	6.1	0.037
0.8	0.8	-2.00	4.59	5.6	0.074
1.0	1.0	0.14	2.34	5.0	0.138

Panel B. Industry-Wide vs. Banking Association Cartel

Null (κ) (1)	Alternative (κ) (2)	T^{RV} (3)	Effective F (4)	Critical Value (5)	ρ^2 (6)
0.2	0.2	3.96	9.44	6.2	0.091
0.4	0.4	4.26	8.03	6.0	0.112
0.6	0.6	4.72	6.68	5.2	0.195
0.8	0.8	3.38	5.10	5.4	0.167
1.0	1.0	2.94	3.47	5.9	0.116

Panel C. Industry-Wide vs. Top 4 Cartel

Null (κ) (1)	Alternative (κ) (2)	T^{RV} (3)	Effective F (4)	Critical Value (5)	ρ^2 (6)
0.2	0.2	3.88	14.38	6.3	0.086
0.4	0.4	4.17	12.44	5.9	0.123
0.6	0.6	4.67	8.21	5.2	0.191
0.8	0.8	2.87	4.21	5.2	0.190
1.0	1.0	1.47	3.15	5.0	0.219